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MIL-M-28787/260A
23 March 1994
SUPERSEDING
MIL-M-28787/260(EC)
3 June 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE EMK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

*The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

* 1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

* AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

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3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.0855 failures per million operating hours for class I or 0.3692 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be KMK.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/260-1 for class I and

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M28787/260-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

* 3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

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TABLE I. Module electrical requirements

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		80° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial End-of-life	III III	S S	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	10 1	INF INF	MOhm MOhm
* B1-B30	Input clamp diode voltage, negative	IV	S	-0.4	-6 0	-0.4	-6.0	-0 4	-6.0	-0.4	-6.0	-0 4	-6 0	V
* C1-C30	Input clamp diode voltage, positive	V	S	0.4	1 5	0.4	1.5	0 4	1.5	0 4	1.5	0 4	1.5	V
D1-D16	Functional test	VI	X	----------				See table VI						----------
E1-E8	Output voltage, low level, IOL = 0	VII	X	0.00	0.05	0 00	0.05	0 00	0.05	0 00	0.05	0 00	0.05	V
F1-F8	Output voltage, high level, IOH = 0	VIII	X	4.95	5 05	4.95	5 05	4 95	5.05	4 95	5.05	4 95	5 05	V
G1-G8	Output voltage, low level, VDD = 5V	IX	X	0 0	0 4	0 0	0 4	0 0	0.4	0.0	0.4	0.0	0 4	V
H1-H8	Output voltage, high level, VDD = 5V	X	X	4.6	5 0	4 6	5 0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J8	Output voltage, low level, VDD = 15V	XI	X	0.0	1 5	0.0	1 5	0.0	1.5	0.0	1.5	0.0	1.5	V
* K1-K8	Output voltage, high level, VDD = 15V	XII	X	13 5	15 15	13.5	15 15	13.5	15.15	13.5	15.15	13.5	15 15	V
L1-L30	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1-M30	Input current, low level	XIV	X	0	-200	---	---	0	-200	---	---	0	-200	nA
N1	Power supply current, low level	XV	X	0	8	0	8	0	78	0	8	0	130	µA
N2	Power supply current, high level	XV	X	0	8	0	8	0	78	0	8	0	130	µA

-See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
* P1-P8	Transition time (tTLH), CL = 50 pF, VDD = 5V	XVI	S	6	200	6	200	7	220	5	150	7	236	nS
* P9-P16	Transition time (tTLH), CL = 150 pF, VDD = 5V	XVI	Q	20	680	20	680	23	750	16	515	24	803	nS
* Q1-Q8	Transition time (tTHL), CL = 50 pF, VDD = 5V	XVII	S	6	200	6	200	7	220	5	150	7	236	nS
* Q9-Q16	Transition time (tTHL), CL = 150 pF, VDD = 5V	XVII	Q	20	680	20	680	23	750	16	515	24	803	nS
* R1-R8	Transition time (tTLH), CL = 50 pF, VDD = 10V	XVIII	Q	3	100	3	100	3	110	2	75	4	118	nS
* S1-S8	Transition time (tTHL), CL = 50 pF, VDD = 10V	XIX	Q	3	100	3	100	3	110	2	75	4	118	nS
* T1-T8	Transition time (tTLH), CL = 50 pF, VDD = 15V	XX	Q	2	80	2	80	3	90	2	60	3	96	nS
* U1-U8	Transition time (tTHL), CL = 50 pF, VDD = 15V	XXI	Q	2	80	2	80	3	90	2	60	3	96	nS
* V1-V30	Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	XXII	S	10	320	10	320	11	355	7	245	11	377	nS
* V31-V60	Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	XXII	Q	16	520	16	520	17	575	12	395	18	613	nS
* W1-W30	Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	XXIII	S	10	320	10	320	11	355	7	245	11	377	nS
* W31-W60	Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	XXIII	Q	16	520	16	520	17	575	12	395	18	613	nS

-See footnotes at end of table.

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TABLE I Module electrical requirements - Continued.

	Test cond	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
							0° ± 5°C		80° ± 5°C		-55° ± 5°C		85° ± 5°C		
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
*	X1-X30	Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	XXIV	Q	4	130	4	130	4	145	3	100	5	154	nS
*	Y1-Y30	Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	XXV	Q	4	130	4	130	4	145	3	100	5	154	nS
*	Z1-Z30	Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	XXVI	Q	3	100	3	100	3	110	2	75	4	118	nS
*	AA1-AA30	Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	XXVII	Q	3	100	3	100	3	110	2	75	4	118	nS
	BB	Power supply filter capacitance	XVIII	S	0.007	0.013	0 007	0 013	0.007	0.013	0.007	0.013	0.007	0.013	µF
	CC1-CC30	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

NOTES:

1/ "X" parameters shall be 100 percent tested during module acceptance tests

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

4/ Unless otherwise specified, limits are for initial and end-of-life

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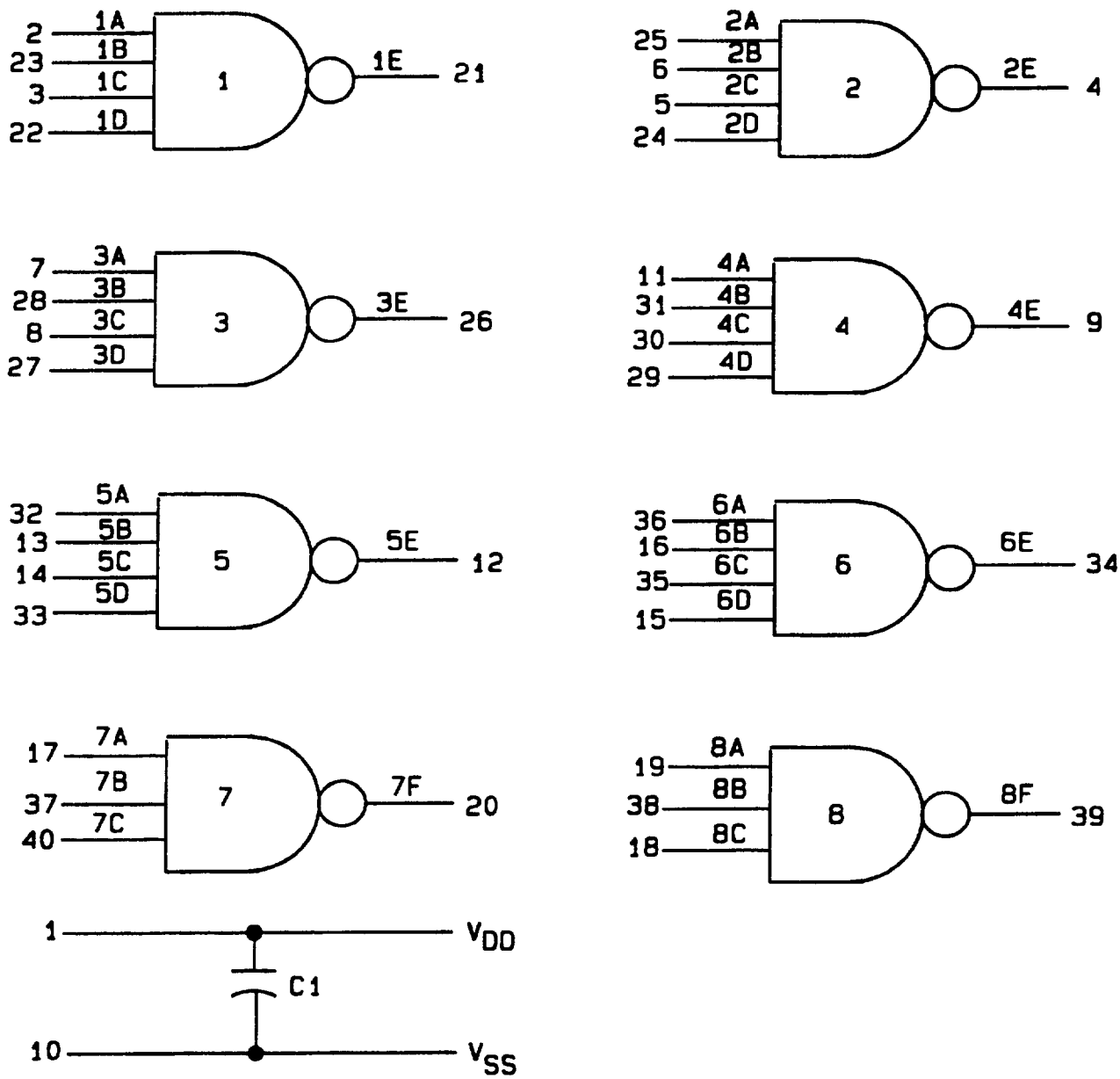


FIGURE 1. **EMK module function diagram, NAND gate.**

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* 3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

* 3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (VDD) of 3 to 18 Vdc ± 5 per cent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B and C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A

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through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through K and N through BB shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions L and M shall be performed at 25°, 60° and 85°C only using the initial limits for the applicable operating temperature.

* 4.5.1.1 Transient thermal test. For the transient thermal tests, use the same test conditions as the maximum class temperature test conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition BB, shall be performed using the initial limits at 25 ± 5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests D through CC shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

5. PACKAGING

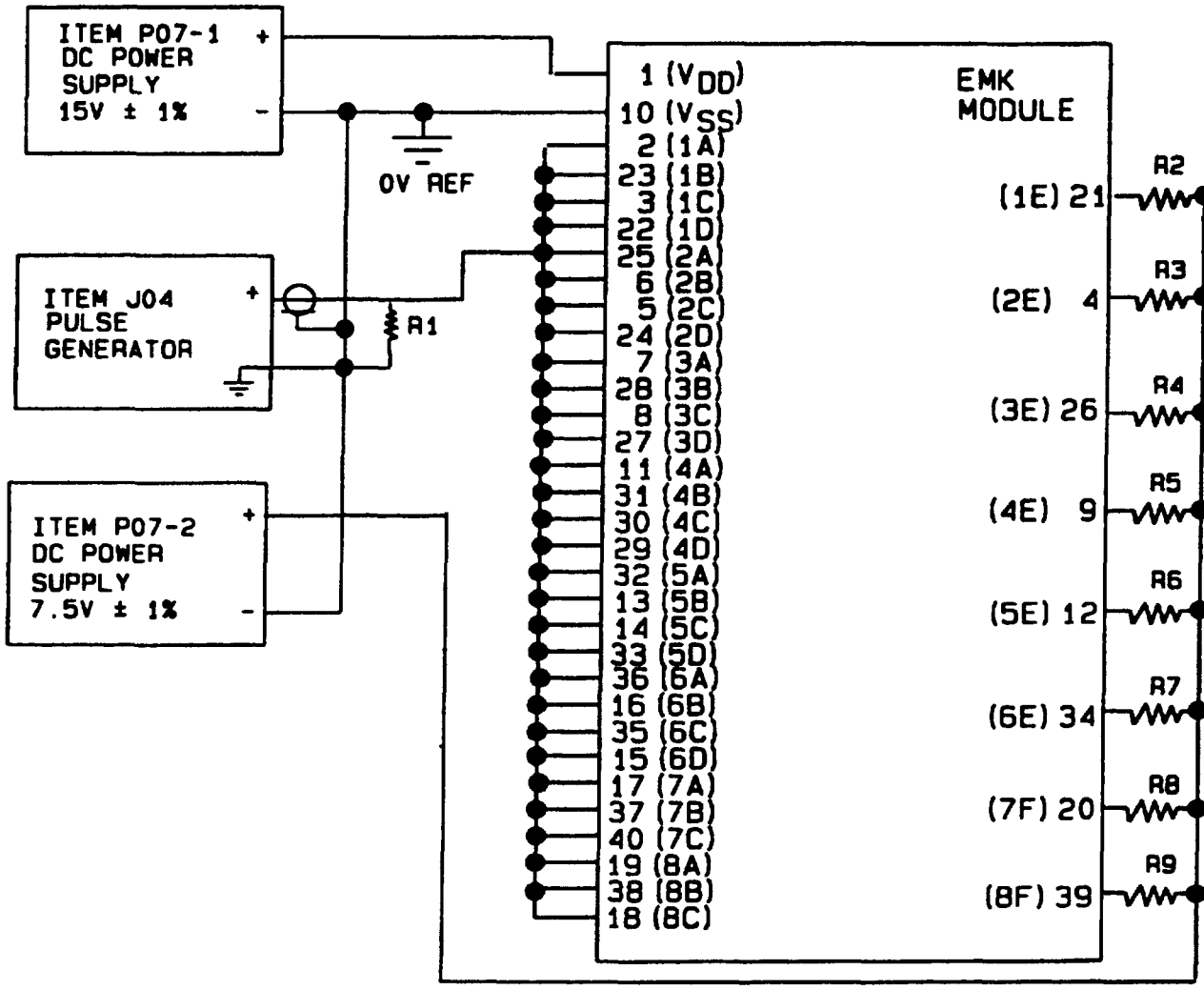
5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

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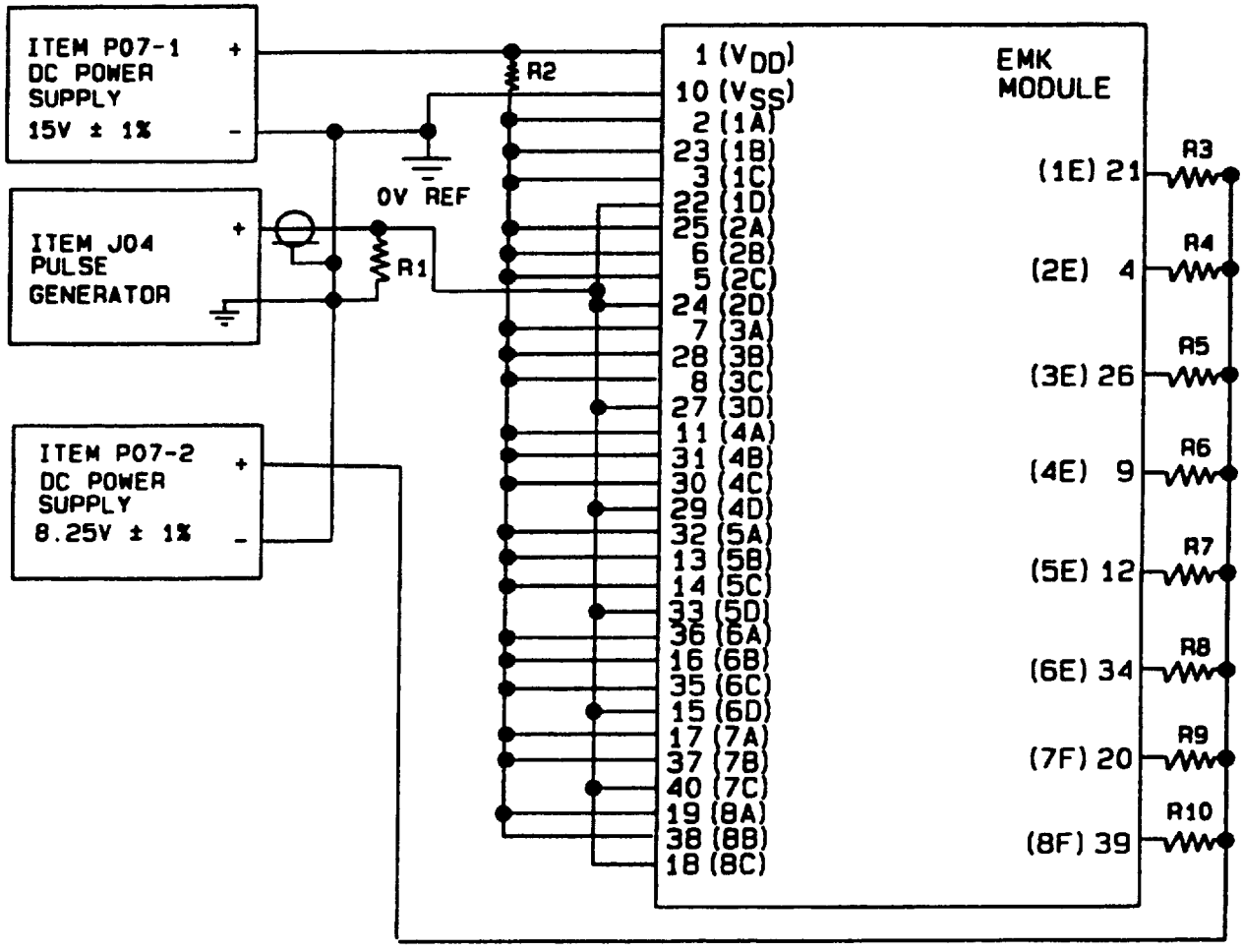


NOTES:

1. R1 = 1.5 kOhms ± 10 per cent, 1 W.
2. R2-R9 = 6.81 kOhms ± 1 per cent, 1/4 W.
3. Item J04 setup:
   - PW = 1,000 ± 100 nS
   - PA = 15 V ± 1 per cent
   - tTLH = tTHL = 50 ± 10 nS
   - Frequency = 0.5 MHz ± 10 per cent

* **FIGURE 2. Life test circuit.**

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NOTES:

1. R1 = 51.1 Ohms ± 5 per cent, 5 W.
2. R2 = 1.5 kOhms ± 1 per cent, 1/4 W.
3. Item J04 setup:
   - PW = 500 ± 100 nS.
   - PA = 15 V ± 1 per cent.
   - tTLH = tTHL = 50 ± 10 nS.
   - Frequency = 1.0 MHz ± 20 per cent.
4. R3 - R10 = 13.7 kOhms ± 1 per cent, 1/4 W.

* FIGURE 3. Component temperature test circuit.

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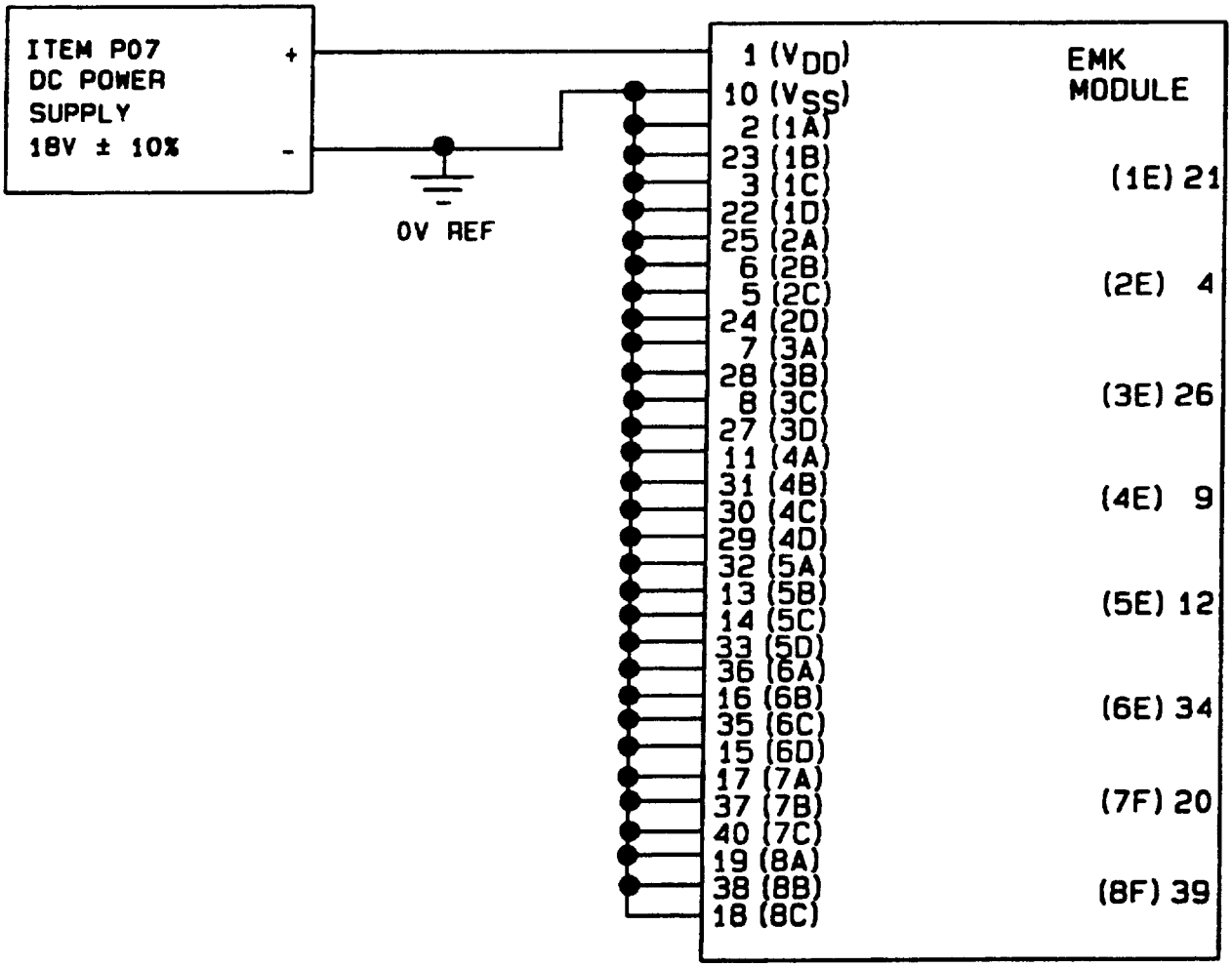


FIGURE 4. Vibration test circuit.

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* 6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life parameters in table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	2.4 max	2.4 max	mW
Input current (VDD = 15V)			
Low level (VIL = 0V)	0 Min, -200 Max	0 Min, -200 Max	nA
Low level (VIL = 15V)	0 Min, 200 Max	0 Min, 200 Max	mA
Output drive current (VDD = 5V)			
Low level (VOL = 0.4V)			
IOL at -55°C	---	0.64 Min	mA
at 0° and 25°C	0.51 Min	0.51 Min	mA
at 60°C	0.44 Min	0.44 Min	mA
at 85°C	---	0.39 Min	mA
High level (VOH = 4.6V)			
IOH at -55°C	---	-0.64 Min	mA
at 0°C and 25°C	-0.51 Min	-0.51 Min	mA
at 60°C	-0.44 Min	-0.44 Min	mA
at 85°C	---	-0.39 Min	mA

6.4 Module function. The module contains six independent four input NAND gates and two independent three input NAND gates.

* 6.4.1 Input to output relationship. Each EMK module has the input to output relationship in table II for positive logic.

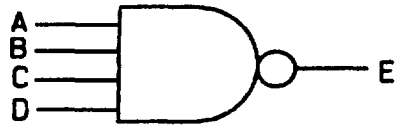
6.4.2 Paralleling of outputs. Paralleling of outputs is permissible when parallel inputs and outputs are confined within each group specified below. The output drive capabilities of this combination are equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

(a) 1 and 2.
(b) 3 and 4.
(c) 5 and 6.
(d) 7 and 8.

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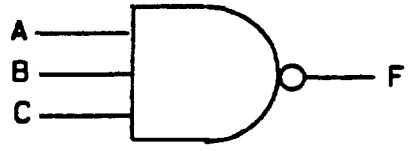
TABLE II. Input to output relationship.

Inputs				Output
A	B	C	D	E
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L



$E = \overline{ABCD}$

Inputs			Output
A	B	C	F
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L



$F = \overline{ABC}$

H = HIGH LEVEL
L = LOW LEVEL

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6.4.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$PTOTAL = (C\ VDD^2\ f)\ N + \text{Quiescent power dissipation.}$$

Where: C = load capacitance.
VDD = Supply voltage.
f = operating frequency.
N = Number of outputs.

6.5 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

(a) Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.

(b) Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

(c) Signals should not be applied to the inputs while the device power supply is off.

(d) All input signals and conditioning voltages must be removed before VDD is disconnected.

(e) Input voltages greater than the supply (VDD - VSS) should not be applied as this will forward bias the protection input diodes.

(f) All unused input leads must be connected to either VSS (GND) or VDD (device supply), whichever is appropriate for the logic circuit involved.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activites:
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Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

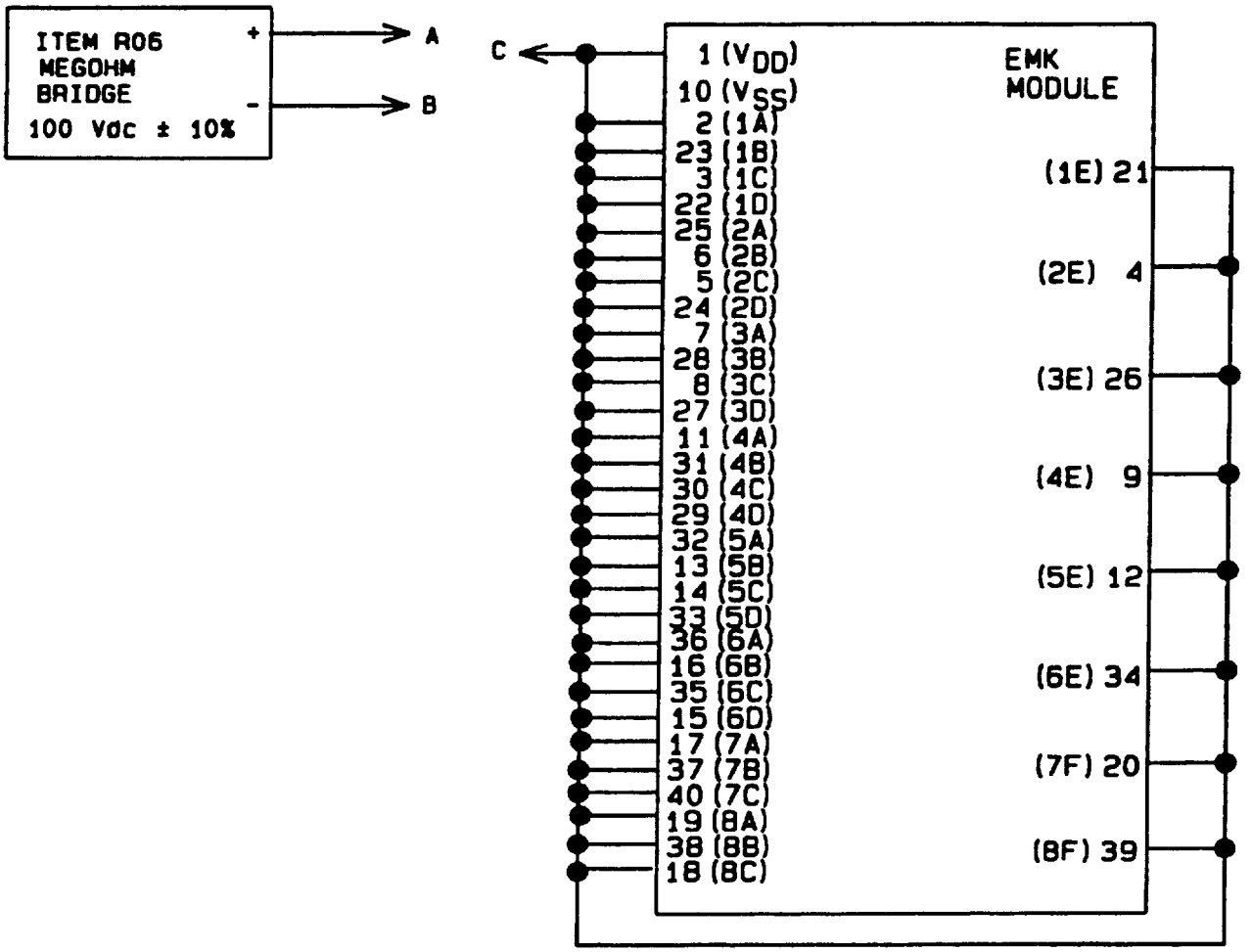
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TABLE III. Isolation test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
A	Isolation	C	Frame	10	INF	MOhm	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

NOTE: When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

* **FIGURE 5. Isolation test circuit for table III.**

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TABLE IV. Input clamp diode voltage (negative) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
B1	1A Input clamp diode voltage, negative	2	Open	-0.4	-6.0	V	6
B2	1B	23					
B3	1C	3					
B4	1D	22					
B5	2A	25					
B6	2B	6					
B7	2C	5					
B8	2D	24					
B9	3A	7					
B10	3B	28					
B11	3C	8					
B12	3D	27					
B13	4A	11					
B14	4B	31					
B15	4C	30					
B16	4D	29					
B17	5A	32					
B18	5B	13					
B19	5C	14					
B20	5D	33					
B21	6A Input clamp diode voltage, negative	36	Open	-0.4	-6.0	V	6

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TABLE IV. Input clamp diode voltage (negative) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
B22	6B Input clamp diode voltage, negative	16	Open	-0.4	-6.0	V	6
B23	6C	35					
B24	6D	15					
B25	7A	17					
B26	7B	37					
B27	7C	40					
B28	8A	19					
B29	8B	38					
B30	8C Input clamp diode voltage, negative	18	Open	-0.4	-6.0	V	6

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TABLE V. Input clamp diode voltage (positive) test.

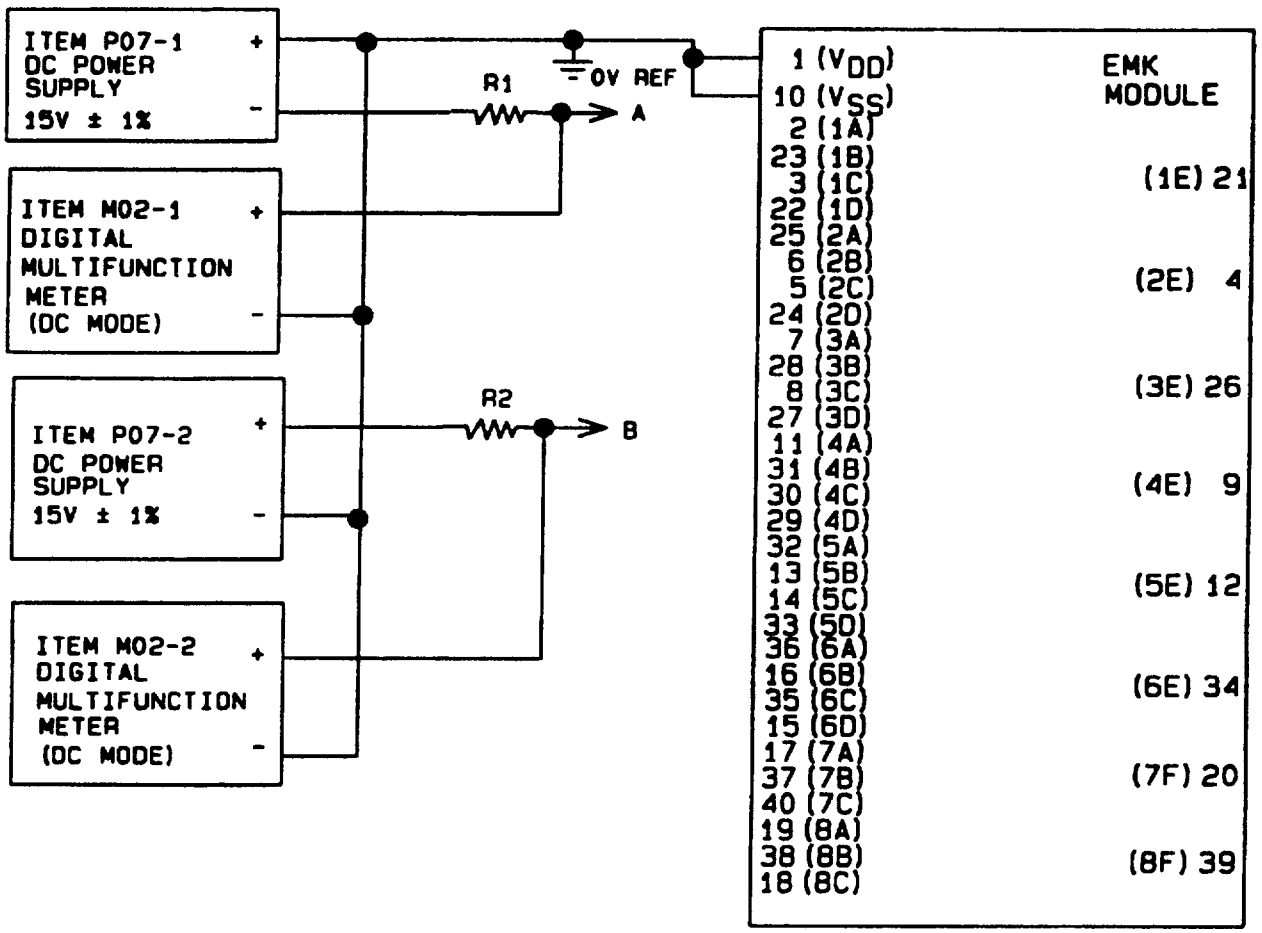
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
C1	1A Input clamp diode voltage, positive	Open	2	0.4	1.5	V	6
C2	1B		23				
C3	1C		3				
C4	1D		22				
C5	2A		25				
C6	2B		6				
C7	2C		5				
C8	2D		24				
C9	3A		7				
C10	3B		28				
C11	3C		8				
C12	3D		27				
C13	4A		11				
C14	4B		31				
C15	4C		30				
C16	4D		29				
C17	5A		32				
C18	5B		13				
C19	5C		14				
C20	5D		33				
C21	6A Input clamp diode voltage, positive	Open	36	0.4	1.5	V	6

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TABLE V. Input clamp diode voltage (positive) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
C22	6B Input clamp diode voltage, positive	Open	16	0.4	1.5	V	6
C23	6C		35				
C24	6D		15				
C25	7A		17				
C26	7B		37				
C27	7C		40				
C28	8A		19				
C29	8B		38				
C30	8C Input clamp diode voltage, positive	Open	18	0.4	1.5	V	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
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NOTES:
1. R1 = 9.09 kOhms ± 1 per cent, 1/8 W.
2. R2 = 13.7 kOhms ± 1 per cent, 1/8 W.
3. Since items P07-1, P07-2, M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

*FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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TABLE VI. Functional test.

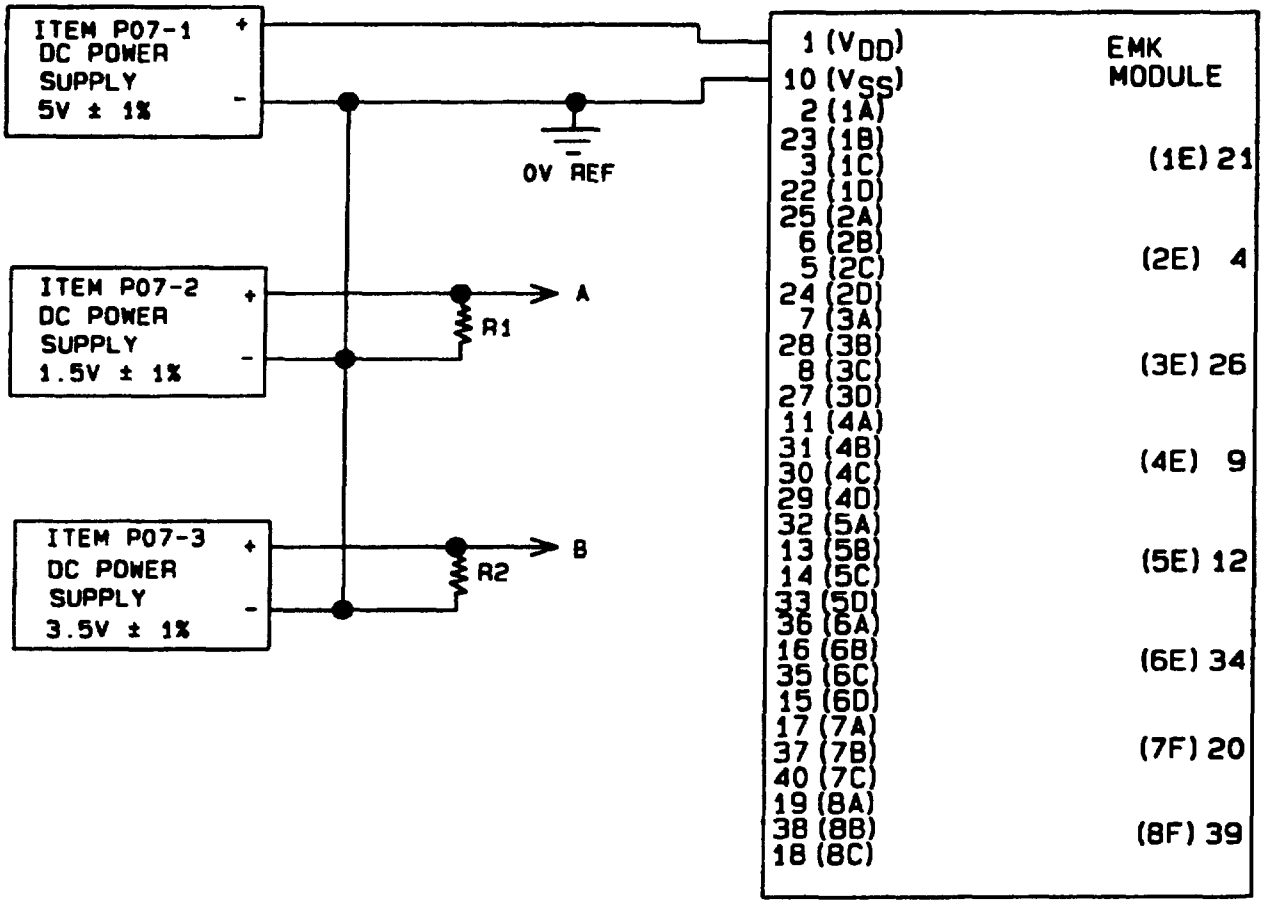
Test cond	Parameter	Connect A to pin	Connect B to pin	Outputs pins 21	4	26	9	12	34	20	39	Test fig
D1	Functional test	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	Open	H	H	H	H	H	H	H	H	7
D2		23,3,22,6,5,24,28,8,27,31,30,29,13,14,33,16,35,15,37,40,38,18	2,25,7,11,32,36,17,19									
D3		2,3,22,25,5,24,7,8,27,11,30,28,32,14,33,36,35,15,17,40,19,18	23,6,28,31,13,16,37,38									
D4		3,22,5,24,8,27,30,29,14,33,35,15,40,18	2,23,25,6,7,28,11,31,32,13,36,16,17,37,19,38									
D5		2,23,22,25,6,24,7,28,27,11,31,29,32,13,33,36,16,15,17,37,19,38	3,5,8,30,14,35,40,18									
D6		23,22,6,24,28,27,31,29,13,33,16,15,37,38	2,3,25,5,7,8,11,30,32,14,36,15,17,40,19,18									
D7		2,22,25,24,7,27,11,29,32,33,36,15,17,19	23,3,6,5,28,8,31,30,13,14,16,35,37,40,38,18							H	H	
D8	Functional test	22,24,27,29,33,15	2,23,3,25,6,5,7,28,8,11,31,30,32,13,14,36,16,35,17,37,40,19,38,18	H	H	H	H	H	H	L	L	7

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* TABLE VI Functional test - Continued

Test cond	Parameter	Connect		Outputs pins								Test fig
		A to pin	B to pin	21	4	26	9	12	34	20	38	
D9	Functional test	2,23,3,25,6,5,7,28,8,11,31,30,32,13,14,36,16,35,17,37,40,19,38,18	22,24,27,29,33,15	H	H	H	H	H	H	H	H	7
D10		23,3,6,5,28,8,31,30,13,14,16,35,17,37,40,19,38,18	2,22,25,24,7,27,11,29,32,33,36,15									
D11		2,3,25,5,7,8,11,30,32,14,36,35,17,37,40,19,38,18	23,22,6,24,28,27,31,29,13,33,16,15									
D12		3,5,8,30,14,35,17,37,40,19,38,18	2,23,22,25,6,24,7,28,27,11,31,29,32,13,33,36,16,15									
D13		2,23,25,6,7,28,11,31,32,13,36,16,17,37,40,19,38,18	3,22,5,24,8,27,30,29,14,33,35,15									
* D14		23,6,28,31,13,16,17,37,40,19,38,18	2,3,22,25,5,24,7,8,27,11,30,29,33,36,35,15,14,32									
D15		2,25,7,11,32,36,17,37,40,19,38,18	23,3,22,8,5,24,28,8,27,31,30,29,13,14,33,16,35,15	H	H	H	H	H	H	H	H	
D16	Functional test	Open	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	L	L	L	L	L	L	L	L	7

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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. High level (H) = 4.6 V max.
3. Low level (L) = 0.4 V max.

FIGURE 7. Functional test circuit for table VI.

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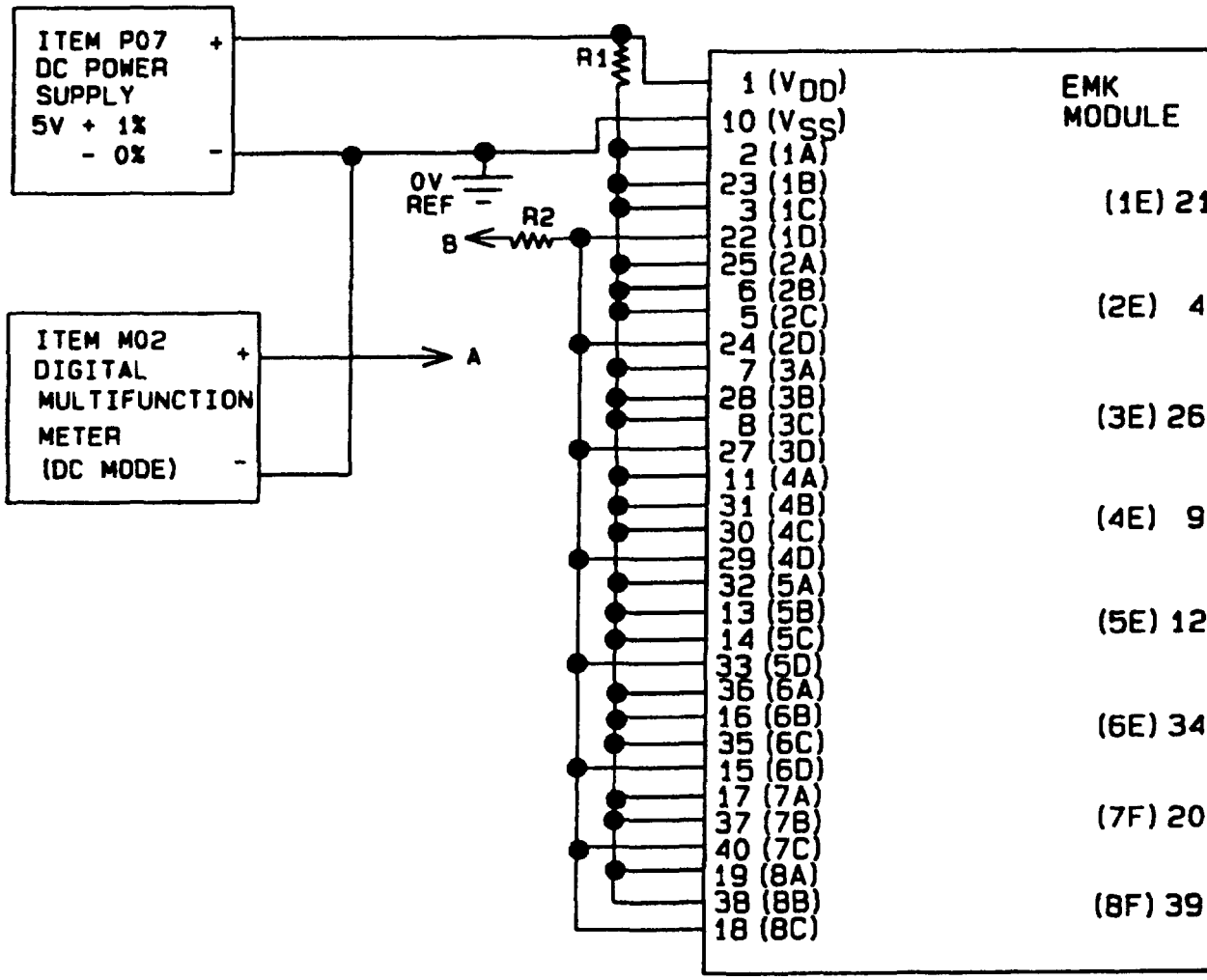
TABLE VII. Output voltage (low level, IOL = 0) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
E1	1E Output voltage, low level, IOL = 0	21	1	0.00	0.05	V	8
E2	2E	4					
E3	3E	26					
E4	4E	9					
E5	5E	12					
E6	6E	34					
E7	7F	20					
E8	8F Output voltage, low level, IOL = 0	39	1	0.00	0.05	V	8

TABLE VIII. Output voltage (high level, IOH = 0) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
F1	1E Output voltage, high level, IOH = 0	21	10	4.95	5.05	V	8
F2	2E	4					
F3	3E	26					
F4	4E	9					
F5	5E	12					
F6	6E	34					
F7	7F	20					
F8	8F Output voltage, high level, IOH = 0	39	10	4.95	5.05	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
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NOTE: R1 = R2 = 1.5 kOhms ± 1 per cent, 1/4 W.

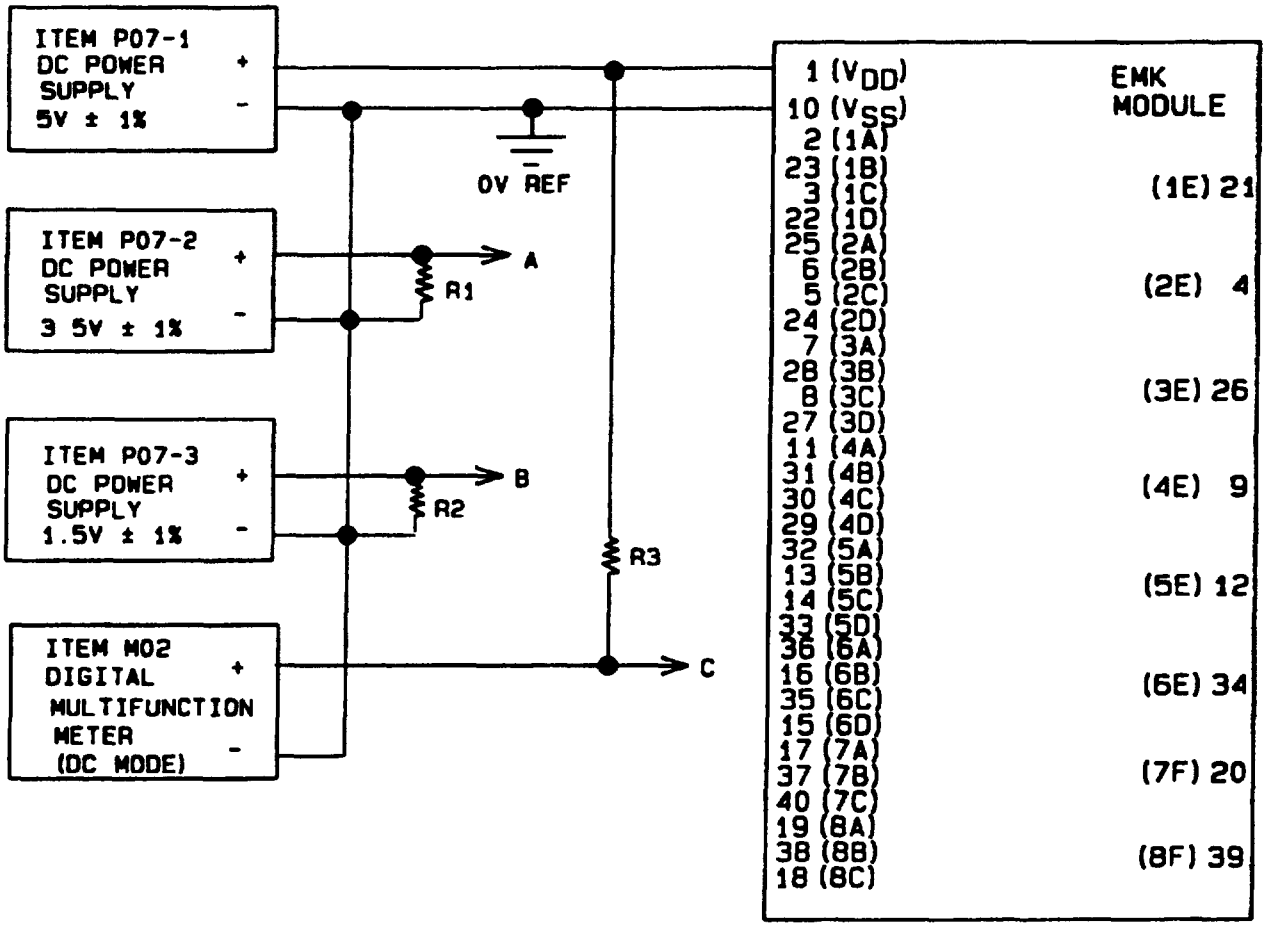
FIGURE 8. Output voltage (low level, IOL = 0 and high level, IOH = 0) test circuit for tables VII and VIII.

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TABLE IX. Output voltage (low level, VDD = 5V) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
G1	1E Output voltage, low level, VDD = 5V	2,23,3,22	25,6,5,24,7,28,8,27, 11,31,30,29,32,13,14, 33,36,16,35,15,17,37, 40,19,38,18	21	0.0	0.4	V	9
G2	2E	25,6,5,24	2,23,3,22,7,28,8,27, 11,31,30,29,32,13,14, 33,36,16,35,15,17,37, 40,19,38,18	4				
G3	3E	7,28,8,27	2,23,3,22,25,6,5,24, 11,31,30,29,32,13,14, 33,36,16,35,15,17,37, 40,19,38,18	26				
G4	4E	11,31,30, 29,	2,23,3,22,25,6,5,24, 7,28,8,27,32,13,14, 33,36,16,35,15,17,37, 40,19,38,18	9				
G5	5E	32,13,14, 33	2,23,3,22,25,6,5,24, 7,28,8,27,11,31,30, 29,36,16,35,15,17,37, 40,19,38,18	12				
G6	6E	36,16,35, 15	2,23,3,22,25,6,5,24, 7,28,8,27,11,31,30, 29,32,13,14,33,17,37, 40,19,38,18	34				
G7	7F	17,37,40	2,23,3,22,25,6,5,24, 7,28,8,27,11,31,30, 29,32,13,14,33,36,16, 35,15,19,38,18	20				
G8	8F Output voltage, low level, VDD = 5V	19,38,18	2,23,3,22,25,6,5,24, 7,28,8,27,11,31,30, 29,32,13,14,33,36,16, 35,15,17,37,40	39	0.0	0.4	V	9

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NOTES:
*1. R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.
*2. R3 = 9.09 kOhms ± 1 per cent, 1/8 W at 0° and 25°C.
= 10.5 kOhms ± 1 per cent, 1/8 W at 60°C.
= 7.15 kOhms ± 1 per cent, 1/8 W at -55°C.
* = 11.8 kOhms ± 1 per cent, 1/8 W at 85°C.

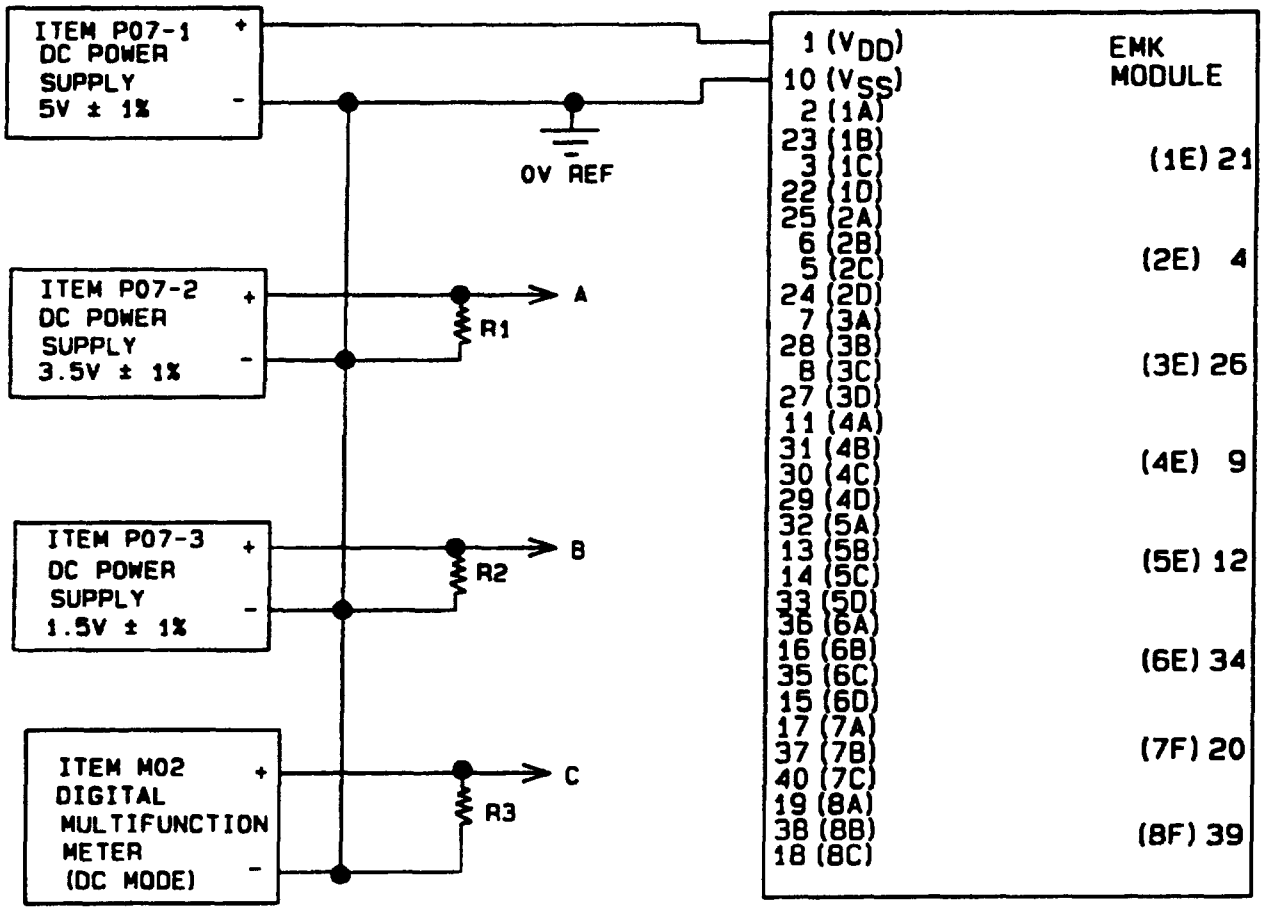
* FIGURE 9. Output voltage (low level, VDD = 5V) test circuit for table IX.

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TABLE X. Output voltage (high level, VDD = 5V) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
H1	1E Output voltage, high level, VDD = 5V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	4.6	5.0	V	10
H2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
H3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
H4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
H5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
H6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
H7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20				
H8	8F Output voltage, high level, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	4.6	5.0	V	10

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NOTES:
1. R1 = R2 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. R3 = 23.2 kOhms ± 1 per cent, 1/8 W at 0° and 25°C.
= 27.4 kOhms ± 1 per cent, 1/8 W at 60°C.
= 18.2 kOhms ± 1 per cent, 1/8 W at -55°C.
= 32.4 kOhms ± 1 per cent, 1/8 W at 85°C.

FIGURE 10. Output voltage (high level, VDD = 5V) test circuit for table X.

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TABLE XI. Output voltage (low level, VDD = 15V) test.

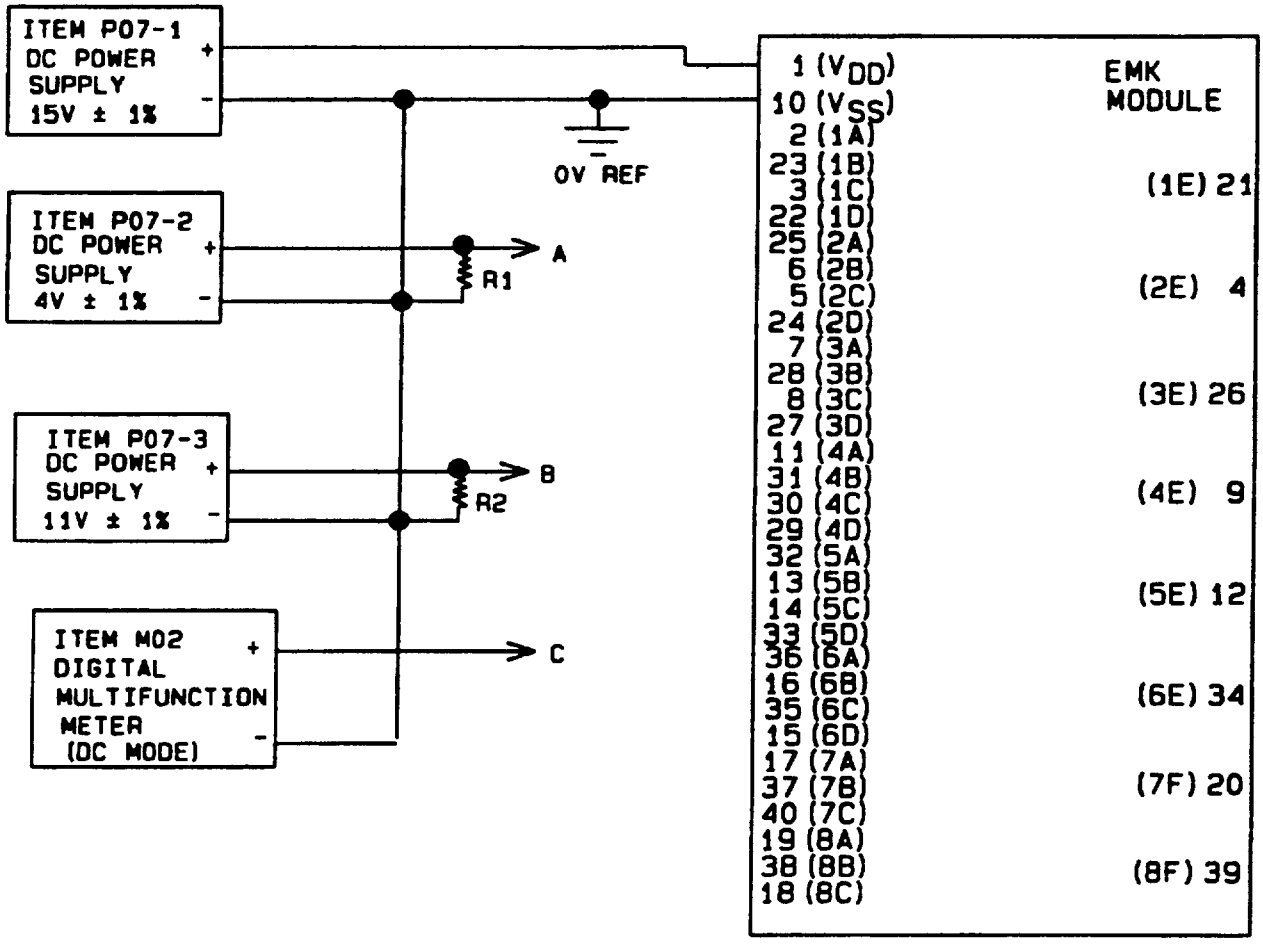
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
J1	1E Output voltage, low level, VDD = 15V	25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2,23,3,22	21	0.0	1.5	V	11
J2	2E	2,23,3,22,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25,6,5,24	4				
J3	3E	2,23,3,22,25,6,5,24,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7,28,8,27	26				
J4	4E	2,23,3,22,25,6,5,24,7,28,8,27,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11,31,30,29	9				
J5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,36,16,35,15,17,37,40,19,38,18	32,13,14,33	12				
J6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,17,37,40,19,38,18	36,16,35,15	34				
J7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,19,38,18	17,37,40	20				
J8	8F Output voltage, low level, VDD = 15V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40	19,38,18	39	0.0	1.5	V	11

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* TABLE XII. Output voltage (high level, VDD = 15V) test.

Test cond		Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
* K1	1E	Output voltage, high level, VDD = 15V	2,23, 3,22	25,6,5,24,7,28,8,27,11, 31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38, 18	21	13.5	15 15	V	11
K2	2E		25,6, 5,24	2,23,3,22,7,28,8,27,11, 31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38, 18	4				
K3	3E		7,28, 8,27	2,23,3,22,25,6,5,24,11, 31,30,29,32,13,14,33,36, 16,35,15,17,37,40,19,38, 18	26				
K4	4E		11,31, 30,29	2,23,3,22,25,6,5,24,7, 28,8,27,32,13,14,33,36, 16,35,15,17,37,40,19,38, 18	9				
K5	5E		32,13, 14,33	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,36, 16,35,15,17,37,40,19,38, 18	12				
K6	6E		36,16, 35,15	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,17,37,40,19,38, 18	34				
K7	7F		17,37, 40	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15,19, 38,18	20				
K8	8F	Output voltage, high level, VDD = 15V	19,38, 18	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15,17, 37,40	39	13.5	15.15	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
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NOTES:
1. R1 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. R2 = 567 Ohms ± 5 per cent, 1/4 W.

FIGURE 11. Output voltage (low level and high level, VDD = 15V) test circuit for tables XI and XII.

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TABLE XIII. Input current (high level) test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L1	1A Input current, high level	2	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	0	200	nA	12
L2	1B	23	2,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18				
L3	1C	3	2,23,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18				
L4	1D	22	2,23,3,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18				
L5	2A	25	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18				
L6	2B	6	2,23,3,22,25,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18				
L7	2C Input current, high level	5	2,23,3,22,25,6,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	0	200	nA	12

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TABLE XIII. Input current (high level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L8	2D Input current, high level	24	2,23,3,22,25,6,5,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	200	nA	12
L9	3A	7	2,23,3,22,25,6,5,24,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L10	3B	28	2,23,3,22,25,6,5,24,7,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L11	3C	8	2,23,3,22,25,6,5,24,7,28, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L12	3D	27	2,23,3,22,25,6,5,24,7,8, 28,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L13	4A	11	2,23,3,22,25,6,5,24,7,28, 8,27,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L14	4B Input current, high level	31	2,23,3,22,25,6,5,24,7,28, 8,27,11,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	200	nA	12

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TABLE XIII. Input current (high level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
L15	4C Input current, high level	30	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	200	nA	12
L16	4D	29	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L17	5A	32	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
L18	5B	13	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,14, 33,36,16,35,15,17,37,40, 19,38,18				
L19	5C	14	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 33,36,16,35,15,17,37,40, 19,38,18				
L20	5D	33	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,36,16,35,15,17,37,40, 19,38,18				
L21	6A Input current, high level	36	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,16,35,15,17,37,40, 19,38,18	0	200	nA	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input current (high level) test - Continued.

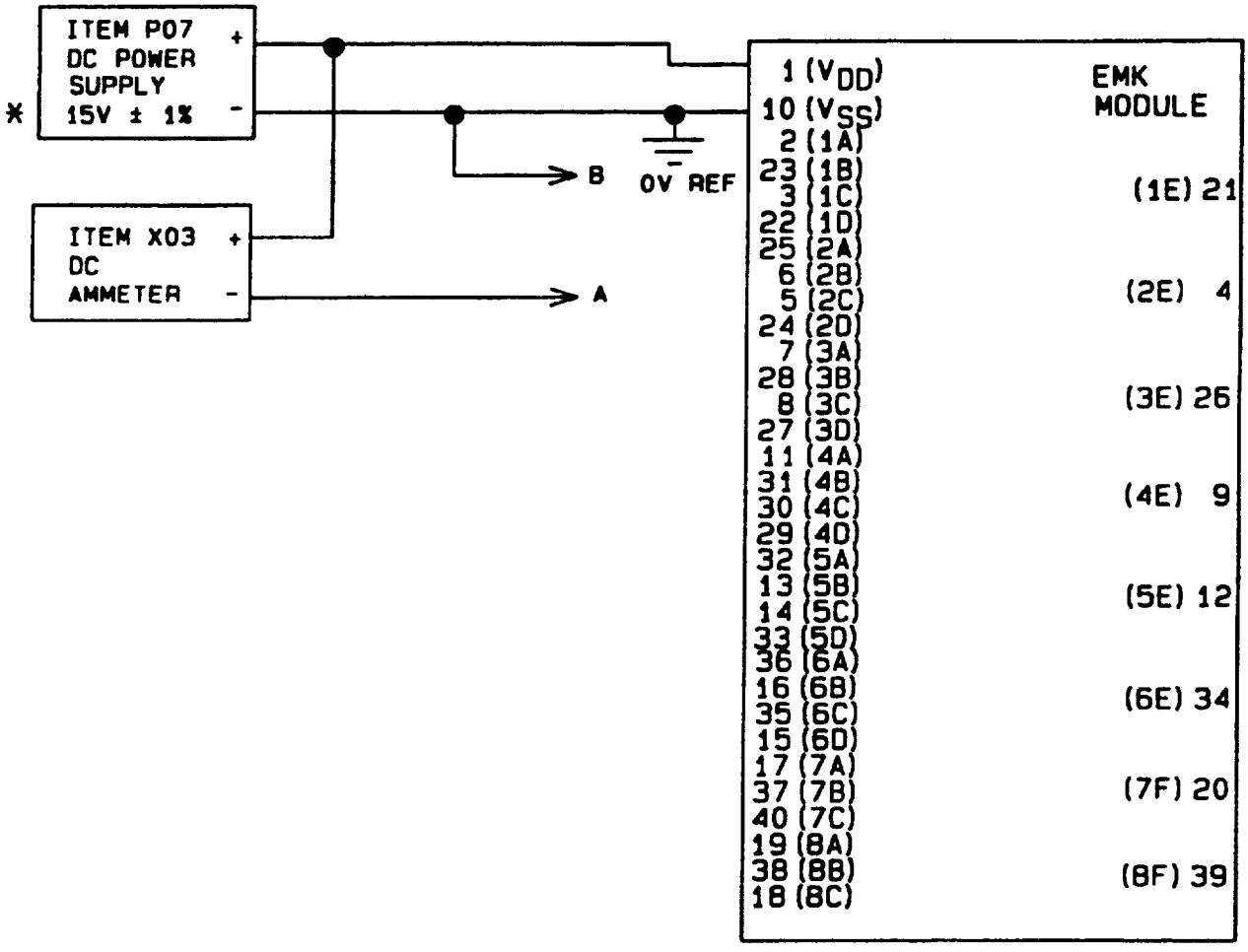
Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L22	6B Input current, high level	16	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,35,15,17,37,40,19,38,18	0	200	nA	12
L23	6C	35	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,15,17,37,40,19,38,18				
L24	6D	15	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,17,37,40,19,38,18				
L25	7A	17	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18				
L26	7B	37	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,40,19,38,18				
L27	7C	40	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,19,38,18				
L28	8A Input current, high level	19	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	0	200	nA	12

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TABLE XIII. Input current (high level) test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L29	8B Input current, high level	38	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,18	0	200	nA	12
L30	8C Input current, high level	18	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38	0	200	nA	12

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CAUTION:
Do not isolate the chassis of ammeter X01 from earth ground.

* **FIGURE 12.** Input current (high level) test for table XIII.

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TABLE XIV Input current (low level) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
M1	1A Input current, low level	2	23,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	-200	nA	13
M2	1B	23	2,3,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M3	1C	3	2,23,22,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M4	1D	22	2,23,3,25,6,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M5	2A	25	2,23,3,22,6,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M6	2B	6	2,23,3,22,25,5,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M7	2C Input current, low level	5	2,23,3,22,25,6,24,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	-200	nA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input current (low level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
M8	2D Input current, low level	24	2,23,3,22,25,6,5,7,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18	0	-200	nA	13
M9	3A	7	2,23,3,22,25,6,5,24,28,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M10	3B	28	2,23,3,22,25,6,5,24,7,8, 27,11,31,30,29,32,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M11	3C	8	2,23,3,22,25,6,5,24,7,27, 11,31,30,29,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18				
M12	3D	27	2,23,3,22,25,6,5,24,7,8, 11,31,30,29,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18				
M13	4A	11	2,23,3,22,25,6,5,24,7,8, 27,31,30,29,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18				
M14	4B Input current, low level	31	2,23,3,22,25,6,5,24,7,8, 27,11,30,29,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input current (low level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
M15	4C Input current, low level	30	2,23,3,22,25,6,5,24,7,8, 27,11,31,29,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18	0	-200	nA	13
M16	4D	29	2,23,3,22,25,6,5,24,7,8, 27,11,31,30,32,13,14,33, 36,16,35,15,17,37,40,19, 38,18				
M17	5A	32	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,13,14, 33,36,16,35,15,17,37,40, 19,38,18				
M18	5B	13	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,14, 33,36,16,35,15,17,37,40, 19,38,18				
M19	5C	14	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 33,36,16,35,15,17,37,40, 19,38,18				
M20	5D	33	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,36,16,35,15,17,37,40, 19,38,18				
M21	6A Input current, low level	36	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,16,35,15,17,37,40, 19,38,18	0	-200	nA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input current (low level) test - Continued.

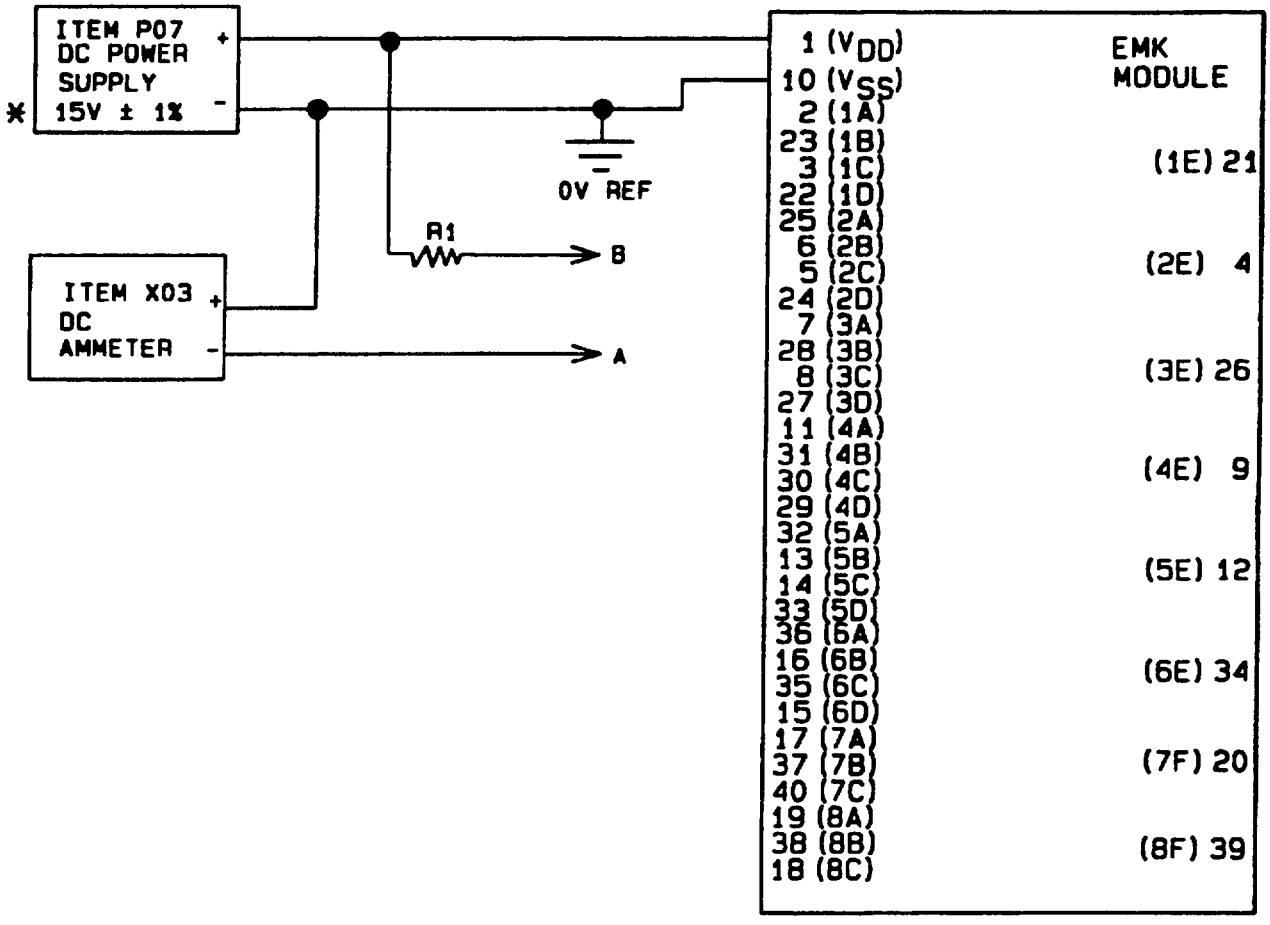
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
M22	6B Input current, low level	16	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,35,15,17,37,40, 19,38,18	0	-200	nA	13
M23	6C	35	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,15,17,37,40, 19,38,18				
M24	6D	15	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,17,37,40, 19,38,18				
M25	7A	17	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,37,40, 19,38,18				
M26	7B	37	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17,40, 19,38,18				
M27	7C	40	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17,37, 19,38,18				
M28	8A Input current, low level	19	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17,37, 40,38,18	0	-200	nA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
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TABLE XIV. Input current (low level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
M29	8B Input current, low level	38	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17,37, 40,19,18	0	-200	nA	13
M30	8C Input current, low level	18	2,23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17,37, 40,19,38	0	-200	nA	13

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CAUTION

Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kOhms ± 5 per cent, 1/4 W.

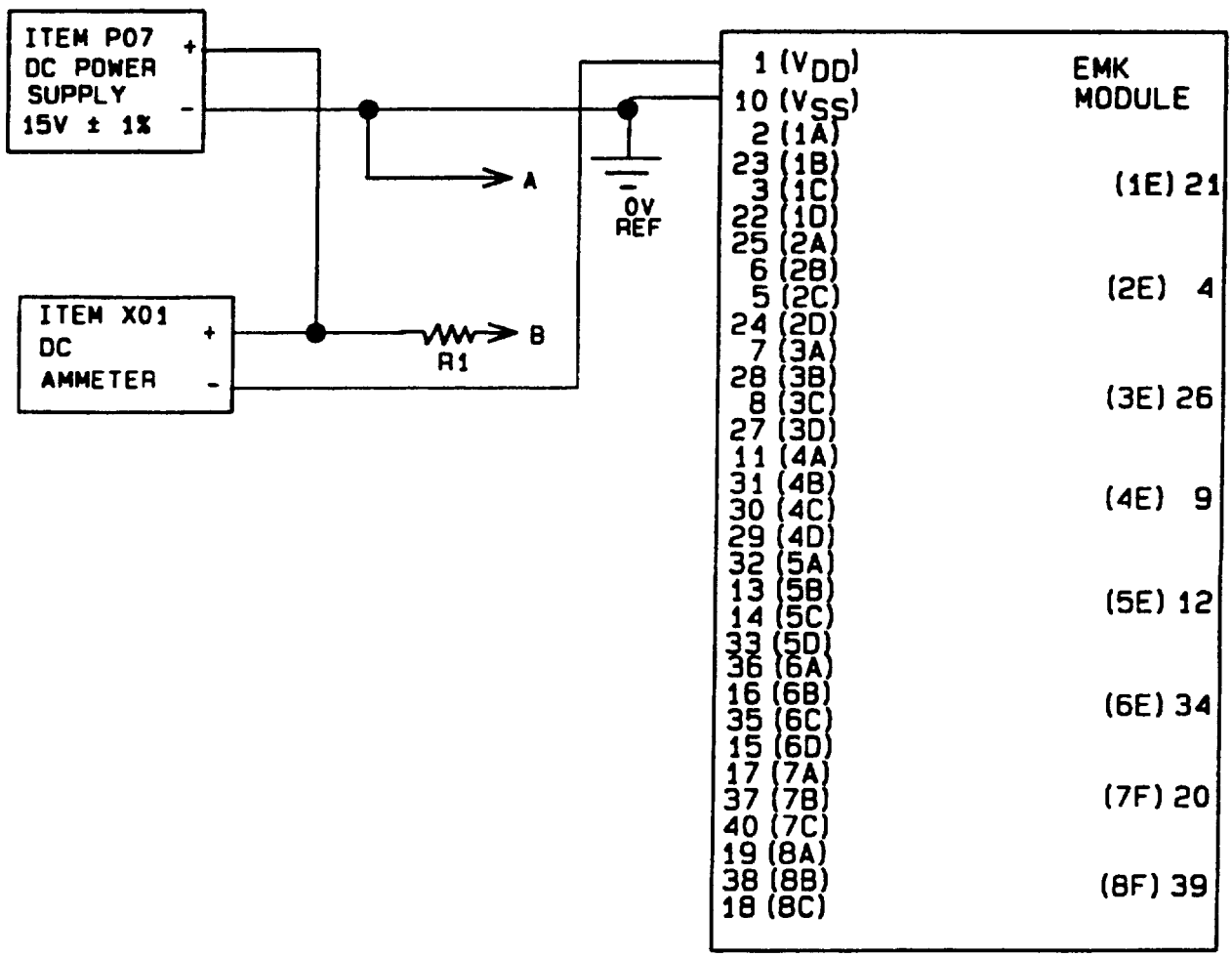
* **FIGURE 13.** Input current (low level) test circuit for table XIV.

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TABLE XV. Power supply current (low level and high level) test.

Test cond	Parameter	Connect		Initial limit 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
N1	Power supply current, low level	Open	2,23,3, 22,25, 6,5,24, 7,28,8, 27,11, 31,30, 29,32, 13,14, 33,36, 16,35, 15,17, 37,40, 19,38, 18	0	8	μA	14
N2	Power supply current, high level	2,23,3, 22,25, 6,5,24, 7,28,8, 27,11, 31,30, 29,32, 13,14, 33,36, 16,35, 15,17, 37,40, 19,38, 18	Open	0	8	μA	14

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CAUTION:
*Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 1.5 kOhms ± 5 per cent, 1/4 W.

* FIGURE 14. Power supply current (low level and high level) test circuit for table XV.

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* TABLE XVI. Transition time (tTLH, VDD = 5V) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
* P1	1E Transition time (tTLH) CL = 50 pF, VDD = 5V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	6	200	nS	15, 17
P2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
P3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
P4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
P5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
P6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
P7	7F Transition time (tTLH) CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20	6	200	nS	15, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Transition time (tTLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
P8	8F Transition time (tTLH) CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15,	19	39	6	200	nS	15, 17
* P9	1E Transition time (tTLH) CL = 150 pF, VDD = 5V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	20	680		
P10	2E	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4				
P11	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
P12	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
P13	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
P14	6E Transition time (tTLH) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	20	680	nS	15, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
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* TABLE XVI. Transition time (tTLH, VDD = 5V) test - Continued

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P15	7F Transition time (tTLH) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20	20	680	nS	15, 17
P16	8F Transition time (tTLH) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	20	680	nS	15, 17

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* TABLE XVII. Transition time (tTHL, VDD = 5V) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
* Q1	1E Transition time (tTHL) CL = 50 pF, VDD = 5V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	6	200	nS	15, 17
Q2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
Q3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
Q4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
Q5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
Q6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
Q7	7F Transition time (tTHL) CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20	6	200	nS	15, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

* TABLE XVII. Transition time (tTHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Q8	8F Transition time (tTHL) CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	6	200	nS	15, 17
* Q9	1E Transition time (tTHL) CL = 150 pF, VDD = 5V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	20	680		
Q10	2E	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4				
Q11	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
Q12	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
Q13	5E Transition time (tTHL) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12	20	680	nS	15, 17

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* TABLE XVII. Transition time (tTHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q14	6E Transition time (tTHL) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	20	680	nS	15, 17
Q15	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,37, 40,19,38,18	17	20				
Q16	8F Transition time (tTHL) CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	20	680	nS	15, 17

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* TABLE XVIII. Transition time (tTLH, VDD = 5V) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
* R1	1E Transition time (tTLH) CL = 50 pF, VDD = 10V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	3	100	nS	16, 17
R2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
R3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
R4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
R5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
R6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
R7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20				
R8	8F Transition time (tTLH) CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	3	100	nS	16, 17

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* TABLE XIX. Transition time (tTHL, VDD = 10V) test

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
* S1	1E Transition time (tTHL) CL = 50 pF, VDD = 10V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	3	100	nS	16, 17
S2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
S3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
S4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
S5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
S6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
S7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20				
S8	8F Transition time (tTHL) CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	3	100	nS	16, 17

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* TABLE XX. Transition time (tTLH, VDD = 15V) test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
* T1	1E Transition time (tTLH) CL = 50 pF, VDD = 15V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	2	80	nS	16, 17
T2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
T3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
T4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
T5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
T6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
T7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20				
T8	8F Transition time (tTLH) CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	2	80	nS	16, 17

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* TABLE XXI. Transition time (tTHL, VDD = 15V) test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
* U1	1E Transition time (tTHL) CL = 50 pF, VDD = 15V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	2	80	nS	16, 17
U2	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
U3	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
U4	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
U5	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
U6	6E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,16,35,15,17,37,40,19,38,18	36	34				
U7	7F	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,37,40,19,38,18	17	20				
U8	8F Transition time (tTHL) CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,38,18	19	39	2	80	nS	16, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
* V1	1E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	10	320	nS	15, 17
V2	1E	2,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	23					
V3	1E	2,23,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	3					
V4	1E	2,23,3,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	22	21				
V5	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
V6	2E	2,23,3,22,25,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	6					
V7	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	5	4	10	320	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
V8	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4	10	320	nS	15, 17
* V9	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
V10	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
V11	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
V12	3E	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26				
V13	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
V14	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31	9	10	320	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
V15	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	10	320	nS	15, 17
V16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* V17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
V18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
V19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
V20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
V21	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	10	320	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
V22	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	10	320	nS	15, 17
V23	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
V24	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* V25	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
V26	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
V27	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
V28	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	10	320	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
V29	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38	39	10	320	nS	15, 17
V30	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	10	320		
* V31	1E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,16,35,15,17,37, 40,19,38,18	2	21	16	520		
V32	1E	2,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	23					
V33	1E	2,23,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	3					
V34	1E	2,23,3,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	22	21				
V35	2E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4	16	520	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
V36	2E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	6	4	16	520	nS	15, 17
V37	2E	2,23,3,22,25,6,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	5					
V38	2E	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4				
* V39	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
V40	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
V41	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
V42	3E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26	16	520	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
V43	4E Propagation delay time (tPLH), CL = 150 pF VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9	16	520	nS	15, 17
V44	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31					
V45	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30					
V46	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29					
* V47	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
V48	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
V49	5E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14	12	16	520	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
V50	5E Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12	16	520	nS	15, 17
V51	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34				
V52	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16					
V53	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
V54	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* V55	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
V56	7F Propagation delay time (tPLH), CL = 150 pF VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37	20	16	520	nS	15, 17

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* TABLE XXII. Propagation delay time (tPLH, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
V57	7F Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20	16	520	nS	15, 17
V58	8F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39				
V59	8F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38					
V60	8F Propagation delay time (tPLH), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	16	520	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
* W1	1E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	10	320	nS	15, 17
W2	1E	2,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	23					
W3	1E	2,23,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	3					
W4	1E	2,23,3,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	22	21				
W5	2E	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4				
W6	2E	2,23,3,22,25,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	6					
W7	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	5	4	10	320	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
W8	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4	10	320	nS	15, 17
* W9	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
W10	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
W11	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
W12	3E	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26				
W13	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
W14	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31	9	10	320	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
W15	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	10	320	nS	15, 17
W16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* W17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
W18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
W19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
W20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,3,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
W21	6E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	10	320	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
W22	6E Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	10	320	nS	15, 17
W23	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
W24	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* W25	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
W26	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
W27	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
W28	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	10	320	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
W29	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38	39	10	320	nS	15,
W30	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	10	320		
* W31	1E Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	16	520		
* W32	1E	2,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	23					
* W33	1E	2,23,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	3					
W34	1E	2,23,3,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	22	21				
W35	2E Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4	16	520	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond		Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	C to pin	Min	Max		
W36	2E	Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	6	4	16	520	nS	15, 17
W37	2E		2,23,3,22,25,6,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	5					
W38	2E		2,23,3,22,25,6,5,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	24					
W39	3E		2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
W40	3E		2,23,3,22,25,6,5,24,7,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	28					
* W41	3E		2,23,3,22,25,6,5,24,7,28,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	8					
W42	3E	Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	27	26	16	520	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
W43	4E Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9	16	520	nS	15, 17
W44	4E	2,23,3,22,25,6,5,24,7,28,8,27,11,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	31					
W45	4E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	30					
W46	4E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,32,13,14,33,36,16,35,15,17,37,40,19,38,18	29	9				
W47	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,13,14,33,36,16,35,15,17,37,40,19,38,18	32	12				
W48	5E	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,14,33,36,16,35,15,17,37,40,19,38,18	13					
* W49	5E Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,33,36,16,35,15,17,37,40,19,38,18	14	12	16	520	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
W50	5E Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12	16	520	nS	15, 17
W51	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34				
W52	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16					
W53	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
W54	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
W55	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
W56	7F Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37	20	16	520	nS	15, 17

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* TABLE XXIII. Propagation delay time (tPHL, VDD = 5V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
* W57	7F Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20	16	520	nS	15, 17
W58	8F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39				
W59	8F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38					
W60	8F Propagation delay time (tPHL), CL = 150 pF, VDD = 5V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	16	520	nS	15, 17

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* TABLE XXIV. Propagation delay time (tPLH, VDD = 10V) test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
* X1	1E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	4	130	nS	16, 17
X2	1E	2,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	23					
X3	1E	2,23,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	3					
X4	1E	2,23,3,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	22	21				
X5	2E	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4				
X6	2E	2,23,3,22,25,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	6					
X7	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	5	4	4	130	nS	16, 17

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• TABLE XXIV. Propagation delay time (tPLH, VDD = 10V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
X8	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4	4	130	nS	16, 17
• X9	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
X10	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
X11	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
X12	3E	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26				
X13	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
X14	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31	9	4	130	nS	16, 17

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* TABLE XXIV. Propagation delay time (tPLH, VDD = 10V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
X15	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	4	130	nS	16, 17
X16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* X17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
X18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
X19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
X20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
X21	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	4	130	nS	16, 17

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* TABLE XXIV. Propagation delay time (tPLH, VDD = 10V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
X22	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	4	130	nS	16, 17
X23	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
X24	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* X25	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
X26	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
X27	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
X28	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	4	130	nS	16, 17

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* TABLE XXIV. Propagation delay time (tPLH, VDD = 10V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
X29	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38	39	4	130	nS	16, 17
X30	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	4	130	nS	16, 17

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* TABLE XXV. Propagation delay time (tPHL, VDD = 10V) test

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
* Y1	1E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	4	130	nS	16, 17
Y2	1E	2,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	23					
Y3	1E	2,23,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	3					
Y4	1E	2,23,3,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	22	21				
Y5	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
Y6	2E	2,23,3,22,25,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	6					
Y7	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	5	4	4	130	nS	16, 17

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* TABLE XXV. Propagation delay time (tPHL, VDD = 10V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Y8	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4	4	130	nS	16, 17
* Y9	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
Y10	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
Y11	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
Y12	3E	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26				
Y13	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
Y14	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31	9	4	130	nS	16, 17

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* TABLE XXV. Propagation delay time (tPHL, VDD = 10V) test - Continued

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
Y15	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	4	130	nS	16, 17
Y16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* Y17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
Y18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
Y19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
Y20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
Y21	6E Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	4	130	nS	16, 17

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* TABLE XXV. Propagation delay time (tPHL, VDD = 10V) test - Continued.

Test cond		Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
Y22	6E	Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	4	130	nS	16, 17
Y23	6E		2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
Y24	6E		2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* Y25	7F		2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
Y26	7F		2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
Y27	7F		2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
Y28	8F	Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	4	130	nS	16, 17

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* TABLE XXV. Propagation delay time (tPHL, VDD = 10V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Y29	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38	39	4	130	nS	16, 17
Y30	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 10V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	4	130	nS	16, 17

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* TABLE XXVI. Propagation delay time (tPLH, VDD = 15V) test.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
* Z1	1E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	23,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	2	21	3	100	nS	16, 17
Z2	1E	2,3,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	23					
Z3	1E	2,23,22,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	3					
Z4	1E	2,23,3,25,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	22	21				
Z5	2E	2,23,3,22,6,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	25	4				
Z6	2E	2,23,3,22,25,5,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	6					
Z7	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,24,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	5	4	3	100	nS	16, 17

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* TABLE XXVI. Propagation delay time (tPLH, VDD = 15V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Z8	2E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,7,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	24	4	3	100	nS	16, 17
* Z9	3E	2,23,3,22,25,6,5,24,28,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	7	26				
Z10	3E	2,23,3,22,25,6,5,24,7,8,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	28					
Z11	3E	2,23,3,22,25,6,5,24,7,28,27,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	8					
Z12	3E	2,23,3,22,25,6,5,24,7,28,8,11,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	27	26				
Z13	4E	2,23,3,22,25,6,5,24,7,28,8,27,31,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	11	9				
Z14	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7,28,8,27,11,30,29,32,13,14,33,36,16,35,15,17,37,40,19,38,18	31	9	3	100	nS	16, 17

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* TABLE XXVI. Propagation delay time (tPLH, VDD = 15V) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
Z15	4E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	3	100	nS	16, 17
Z16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* Z17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
Z18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
Z19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
Z20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
Z21	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	3	100	nS	16, 17

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* TABLE XXVI. Propagation delay time (tPLH, VDD = 15V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
Z22	6E Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	3	100	nS	16, 17
Z23	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
Z24	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* Z25	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
Z26	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
Z27	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
Z28	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	3	100	nS	16, 17

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* TABLE XXVI. Propagation delay time (tPLH, VDD = 15V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Z29	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,10,35,15, 17,37,40,19,18	38	39	3	100	nS	16, 17
Z30	8F Propagation delay time (tPLH), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,10,35,15, 17,37,40,19,38	18	39	3	100	nS	16, 17

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* TABLE XXVII. Propagation delay time (tPHL, VDD = 15V) test.

	Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ± 5°C ambient Min	Initial limits 25° ± 5°C ambient Max	Unit	Test fig
*	AA1	1E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	23,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	2	21	3	100	nS	16, 17
	AA2	1E	2,3,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	23					
	AA3	1E	2,23,22,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	3					
	AA4	1E	2,23,3,25,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	22	21				
	AA5	2E	2,23,3,22,6,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	25	4				
	AA6	2E	2,23,3,22,25,5,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	6					
	AA7	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,24,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	5	4	3	100	nS	16, 17

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* TABLE XXVII. Propagation delay time (tPHL, VDD = 15V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
AA8	2E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,7,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	24	4	3	100	nS	16, 17
* AA9	3E	2,23,3,22,25,6,5,24,28, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	7	26				
AA10	3E	2,23,3,22,25,6,5,24,7, 8,27,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	28					
AA11	3E	2,23,3,22,25,6,5,24,7, 28,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38,18	8					
AA12	3E	2,23,3,22,25,6,5,24,7, 28,8,11,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	27	26				
AA13	4E	2,23,3,22,25,6,5,24,7, 28,8,27,31,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	11	9				
AA14	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,30,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	31	9	3	100	nS	16, 17

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* TABLE XXVII. Propagation delay time (tPHL, VDD = 15V) test - Continued.

Test cond	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ± 5°C ambient: Min	Initial limits 25° ± 5°C ambient: Max	Unit	Test fig
AA15	4E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,29,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	30	9	3	100	nS	16, 17
AA16	4E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,32,13, 14,33,36,16,35,15,17, 37,40,19,38,18	29	9				
* AA17	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,13, 14,33,36,16,35,15,17, 37,40,19,38,18	32	12				
AA18	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 14,33,36,16,35,15,17, 37,40,19,38,18	13					
AA19	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,33,36,16,35,15,17, 37,40,19,38,18	14					
AA20	5E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,36,16,35,15,17, 37,40,19,38,18	33	12				
AA21	6E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,16,35,15,17, 37,40,19,38,18	36	34	3	100	nS	16, 17

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* TABLE XXVII. Propagation delay time (tPHL, VDD = 15V) test - Continued.

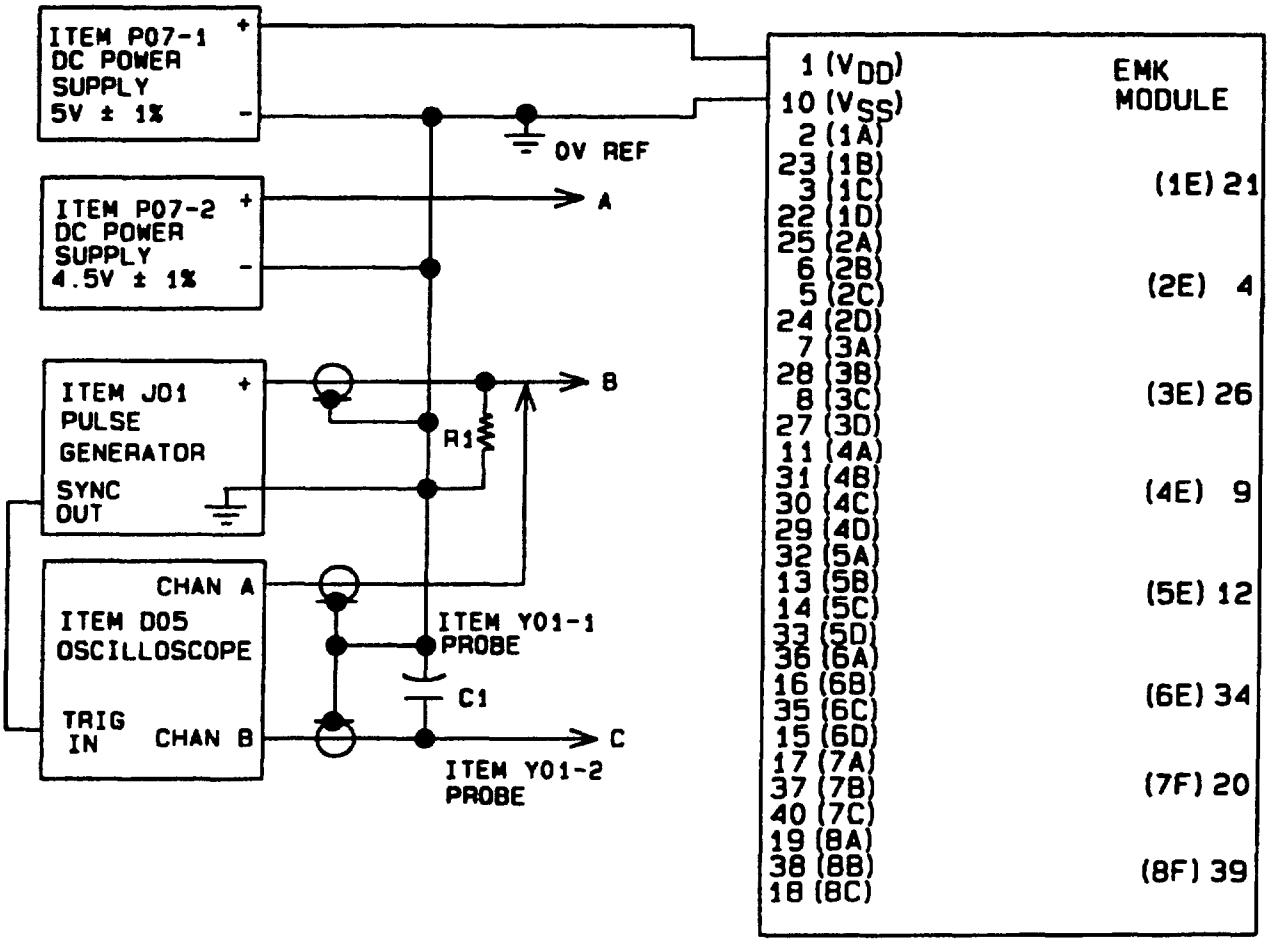
Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
AA22	6E Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,35,15,17, 37,40,19,38,18	16	34	3	100	nS	16, 17
AA23	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,15,17, 37,40,19,38,18	35					
AA24	6E	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,17, 37,40,19,38,18	15	34				
* AA25	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 37,40,19,38,18	17	20				
AA26	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,40,19,38,18	37					
AA27	7F	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,19,38,18	40	20				
AA28	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,38,18	19	39	3	100	nS	16, 17

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* TABLE XXVII. Propagation delay time (tPHL, VDD = 15V) test - Continued.

Test cond	Parameter	Connect			Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
AA29	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,18	38	39	3	100	nS	16, 17
AA30	8F Propagation delay time (tPHL), CL = 50 pF, VDD = 15V	2,23,3,22,25,6,5,24,7, 28,8,27,11,31,30,29,32, 13,14,33,36,16,35,15, 17,37,40,19,38	18	39	3	100	nS	16, 17

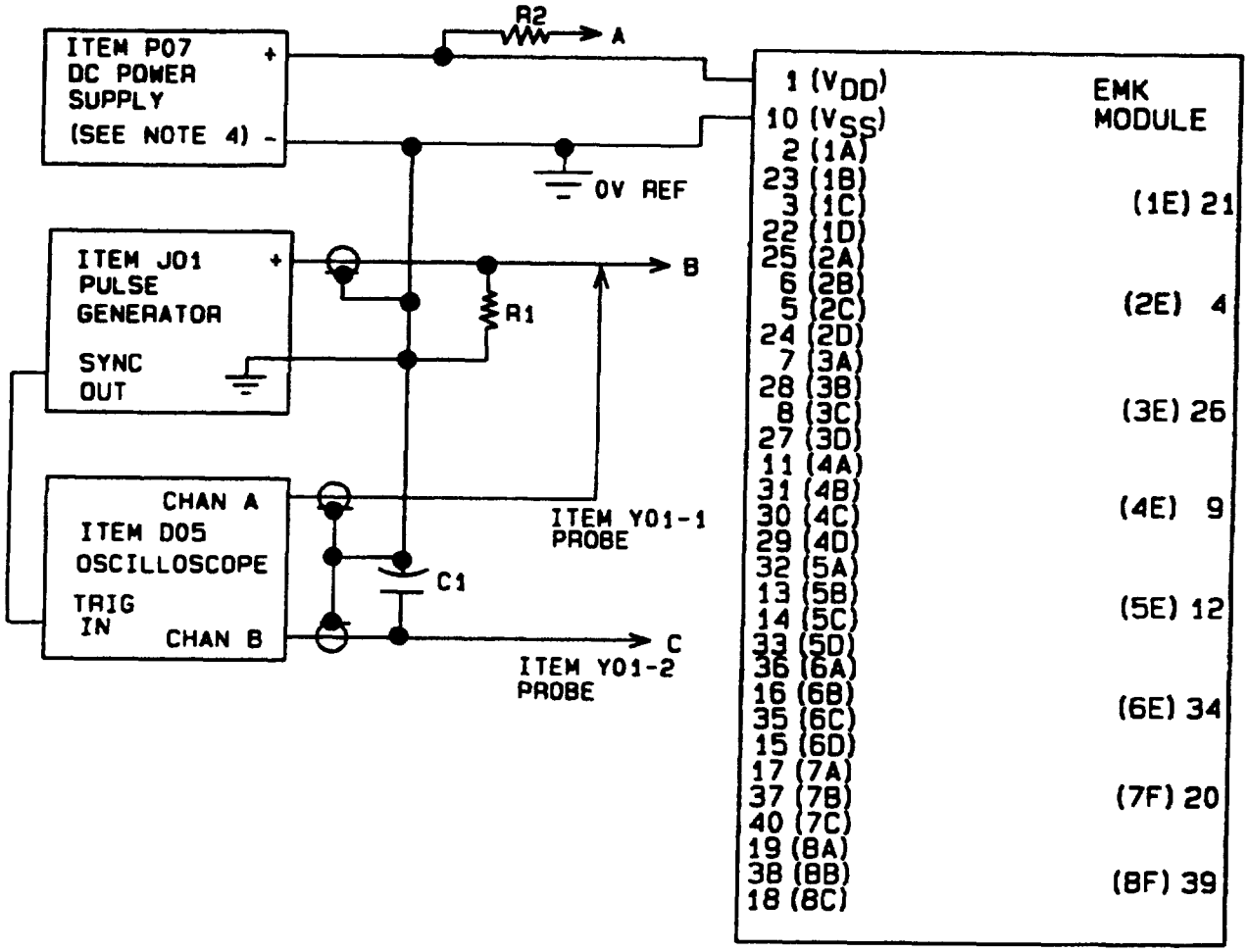
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NOTES:
1. R1 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. C1 = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions P1-P8, Q1-Q8, V1-V30 and W1-W30.
   = 150 ± 15 pF (including probe and parasitic capacitance) for test conditions P9-P16, Q9-Q16, V31-V60 and W31-W60.

FIGURE 15. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XVI, XVII, XXII, and XXIII.

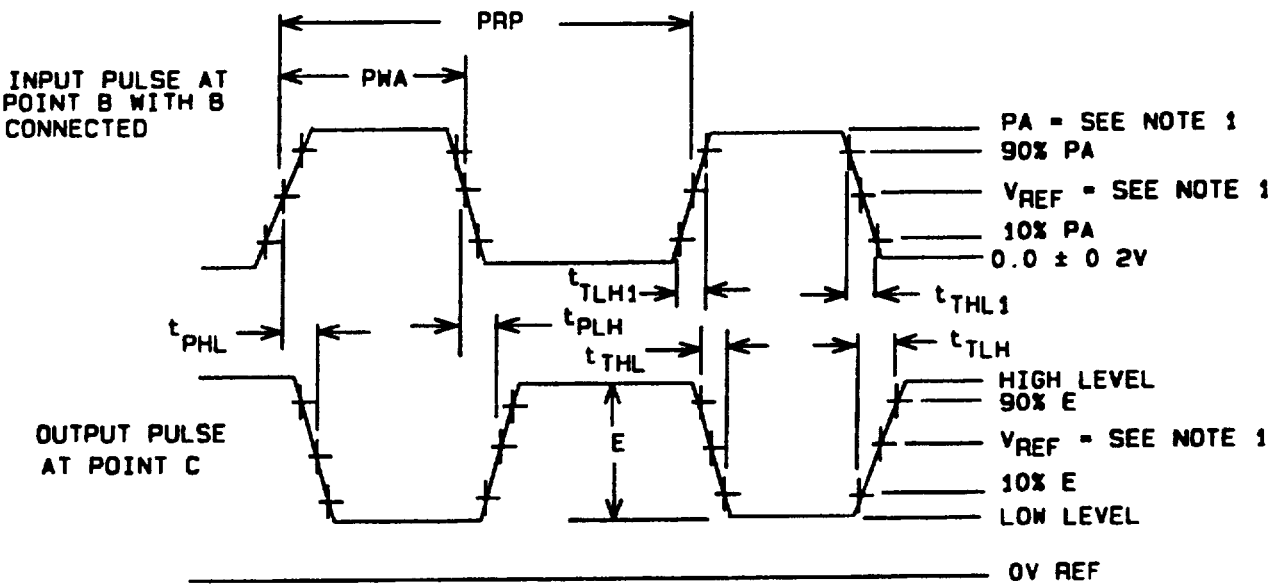
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NOTES:
1. R1 = 51.1 Ohms ± 5 per cent, 1/2 W.
2. R2 = 1.5 kOhms ± 1 per cent, 1/8 W.
3. C1 = 50 ± 5 pF (including probe and parasitic capacitance).
*4. Set item P07 to 10V ± 1 per cent for test conditions R1-R8, S1-S8, X1-X30, and Y1-Y30 and 15V ± 1 per cent for test conditions T1-T8, U1-U8, Z1-Z30 and AA1-AA30.

* FIGURE 16. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XVIII, XIX, XX, XXI, XXIV, XXV, XXVI, and XXVII.

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* FIGURE 17. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit waveforms for tables XVI through XXVII.

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NOTES:
1. Input pulse parameters:
   PA = 5.0 ± 0.1 V for test conditions P1-P16, Q1-Q16, V1-V60 and W1-W60.
      = 10.0 ± 0.1 V for test conditions R1-R8, S1-S8, X1-X30 and Y1-Y30.
      = 15.0 ± 0.1 V for test conditions T1-T8, U1-U8, Z1-Z30 and AA1-AA30.
   VREF = 2.5 ± 0.1 V for test conditions P1-P16, Q1-Q16, V1-V60 and W1-W60.
      = 5.0 ± 0.1 V for test conditions R1-R8, S1-S8, X1-X30 and Y1-Y30.
      = 7.5 ± 0.1 V for test conditions T1-T8, U1-U8, Z1-Z30 and AA1-AA30.
   PRP = 5.0 ± 0.5 µS.
   PWA = 2.5 ± 0.2 µS for test conditions P1-P16, Q1-Q16, V1-V60 and W1-W60.
      = 1.0 ± 0.1 µS for test conditions R1-R8, S1-S8, T1-T8,
   * U1-U8, X1-X30, Y1-Y30, Z1-Z30 and AA1-AA30.
   tTLH = tTHL = 10 ± 2 nS.
2. Before making the measurements specified, verify that the output pulse high level and low level is as follows:
   High level ≥ 4.6 V for test conditions P1-P16, Q1-Q16, V1-V60 and W1-W60.
   ≥ 9.5 V for test conditions R1-R8, S1-S8, X1-X30 and Y1-Y30.
   ≥ 13.5 V for test conditions T1-T8, U1-U8, Z1-Z30 and AA1-AA30.
   Low level ≤ 0.4 V for test conditions P1-P16, Q1-Q16, V1-V60 and W1-W60.
   ≤ 0.5 V for test conditions R1-R8, S1-S8, X1-X30 and Y1-Y30.
   ≤ 1.5 V for test conditions T1-T8, U1-U8, Z1-Z30 and AA1-AA30.
3. The lead length of electrical conductors between terminator resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins and between load circuit and output pin shall be less than one inch (25.4 mm).
4. All voltages are referenced to VSS.

* FIGURE 17. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit waveforms for tables XVI through XXVII - Continued.

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TABLE XXVIII. Power supply filter capacitance test.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
BB	Power supply filter capacitance	Open	1	0.007	0.013	μF	18

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TABLE XXIX. Input capacitance test.

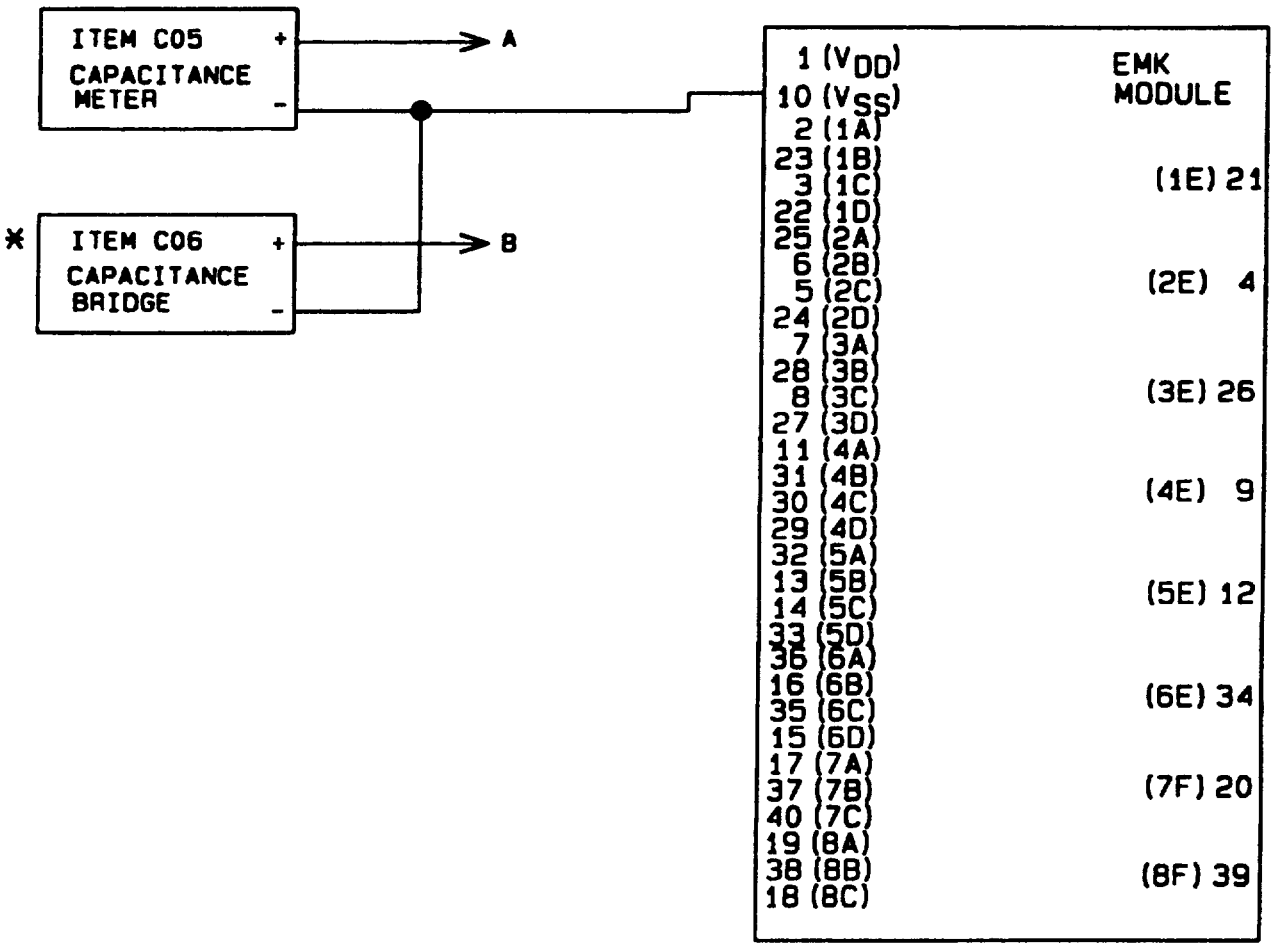
Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
CC1	1A Input capacitance	2	Open	2	15	pF	18
CC2	1B	23					
CC3	1C	3					
CC4	1D	22					
CC5	2A	25					
CC6	2B	6					
CC7	2C	5					
CC8	2D	24					
CC9	3A	7					
CC10	3B	28					
CC11	3C	8					
CC12	3D	27					
CC13	4A	11					
CC14	4B	31					
CC15	4C	30					
CC16	4D	29					
CC17	5A	32					
CC18	5B	13					
CC19	5C	14					
CC20	5D	33					
CC21	6A Input capacitance	36	Open	2	15	pF	18

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TABLE XXIX. Input capacitance test - Continued.

Test cond	Parameter	Connect		Initial limits 25° ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
CC22	6B Input capacitance	16	Open	2	15	pF	18
CC23	6C	35					
CC24	6D	15					
CC25	7A	17					
CC26	7B	37					
CC27	7C	40					
CC28	8A	19					
CC29	8B	38					
CC30	8C Input capacitance	18	Open	2	15	pF	18

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
*3. Capacitance bridge (C06) setup:
Frequency = 1.0 kHz.
Vpk = 0.3 V maximum.

* FIGURE 18. Power supply filter capacitance and input capacitance test circuit for tables XXVIII and XXIX.

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